Chapter 6

Bipolar junction Transistors

6.1 Introduction

The Bipolar Junction Transistor (BJT) is one of the most important solid-state devices. It is a two-junction, three-terminal device, in which the current flow properties of one p-n junction can be modulated by another p-n junction. The structure can be either p-n-p or n-p-n. The three terminals are called emitter, base and collector. The circuit symbols of the two types of BJT are shown in Fig.6.1. The arrow on the emitter signifies the direction of the current flowing at the emitter terminal. In most of our following discussions we shall focus on the n-p-n transistor, which is the device predominantly used in integrated circuits. However, the analysis will be valid for the p-n-p transistor as well, with appropriate changes in carrier types and directions of currents.

6.2 Principle of Operation

In order to understand the interaction of the two p-n junctions in a BJT, let us first consider the structure shown in Fig.6.2(a) in which a region of p-silicon is sandwiched between two n regions (n_1 and n_2). Let us assume that the thickness of the intermediate p-layer is much larger than the diffusion length of electrons (L_n). If one of the two junctions (J_1) is now forward biased and the other junction (J_2) is reverse-biased, practically all the electrons injected by the forward-biased junction into the p-region will recombine before they reach the second junction. The electron current at J_2, which is proportional to the slope of the electron concentration profile, is almost zero. Thus the two junctions are independent of each other. No appreciable current flows at the n_2 terminal and the structure is equivalent to two diodes connected back to back.

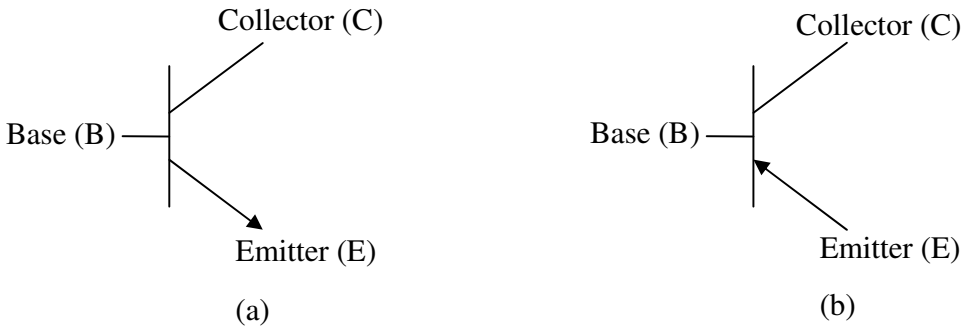


Fig.6.1: Circuit symbol of (a) npn and (b) pnp transistor

However, as shown in Fig. 6.2(b), if the width of the p-region (base) is very small (less than L_n), most of the electrons injected into the p-region from the n_1-region (emitter) reaches J_2. These electrons are then swept across the depletion region of this reverse-biased junction by the large electric field present and are finally collected in the n_2-region (collector). So, even though J_2 is reverse biased, a large current flows at the n_2 terminal due to the proximity of the two junctions. This is the principal feature of transistor action. The n_1-region emitting electrons when J_1 is forward biased, is called the emitter and the narrow p-region is called the base of the transistor.

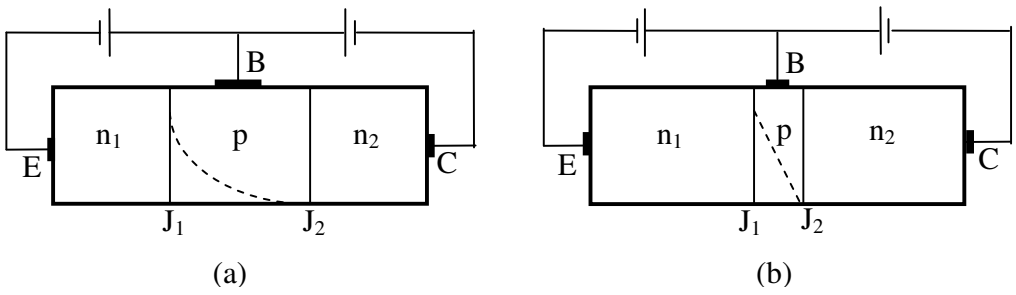


Fig.6.2: n-p-n structures where (a) the width of the p-region is much greater than the diffusion length of electrons and (b) the width of the p-region is much less than the diffusion length of electrons. The electron concentration profile in the p-region (base) is also shown.

6.3 Current Components in a BJT

Fig.6.3 shows a one-dimensional n-p-n transistor structure, which we shall use for analysis. As shown in the figure, the emitter, base and collector widths are W_E, W_B and W_C respectively. The neutral base extends from x = 0 to x = W_B, while the neutral emitter and collector lies at $-(x_E + W_E) < x < -x_E$ and $x_C < x < (x_C + W_C)$ respectively. The doping concentrations in the emitter, base and collector are N_{DE}, N_A and N_{DC} respectively. The transistor has a cross-sectional area 'A', and is uniform along the cross-section. The applied voltage across the emitter-base junction is denoted by V_{BE}, while the applied voltage across the collector-base junction is V_{BC}. The assumed directions of flow of the terminal currents, I_E, I_B and I_C are also shown in the figure.

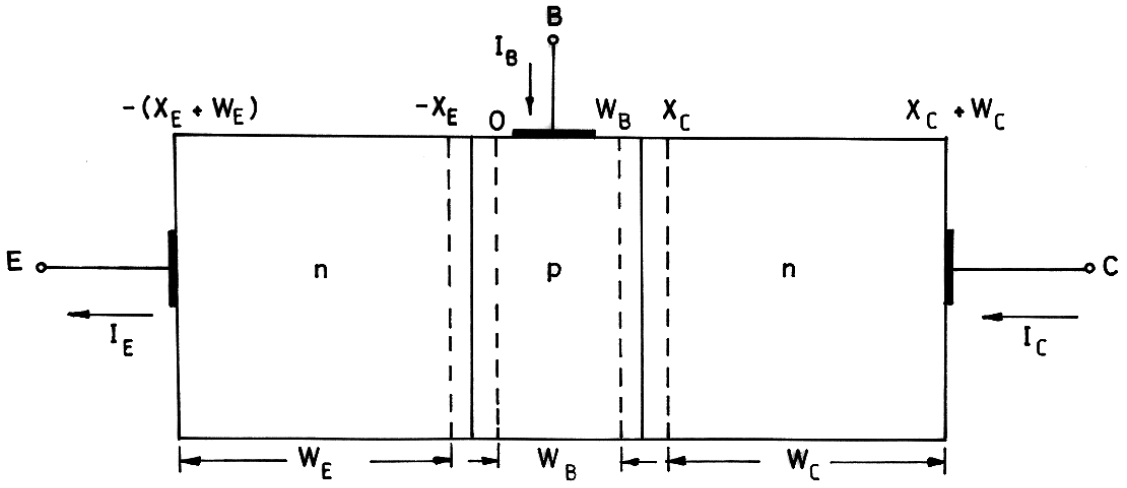


Fig. 6.3: One-dimensional schematic diagram of an npn transistor explaining the various symbols used in text

Since the transistor has two junctions, viz. emitter-base (E-B) and collector-base (C-B), and each of these junctions can be either forward or reverse biased, we can have four different modes of operation for the bipolar transistor. They are:

(a) **Normal active mode** – E-B junction is forward biased and C-B junction is reverse biased,

(b) **Saturation mode** – Both E-B and C-B junctions are forward biased,

(c) **Inverse active mode** – E-B junction is reverse biased and C-B junction is forward biased, and

(d) **Cutoff mode** – Both E-B and C-B junctions are reverse biased.

In order to obtain expressions for the terminal currents, we shall use the minority carrier profiles and currents in the neutral regions. This is because, for minority carriers the drift current can be neglected, as we assume that all the applied voltages appear across the space charge region and there is negligible electric field in the neutral regions. The continuity equations are therefore simplified. We have already seen in Chapter 4 that under bias, the minority carrier concentration at the edge of the depletion region gets modified by a factor $\exp\left(\frac{V}{V_T}\right)$, where $V = V_f$ for a forward biased junction and $V = -V_r$ for a reverse biased junction. Also, at the emitter and collector contacts, the large surface recombination velocity reduces the minority carrier concentration to the thermal equilibrium values. Using these considerations, the minority carrier concentration profiles in the BJT for the different modes of operation will be as shown in Fig. 6.4. The polarities of V_{BE} and V_{BC} for each mode of operation are also shown in the figure.

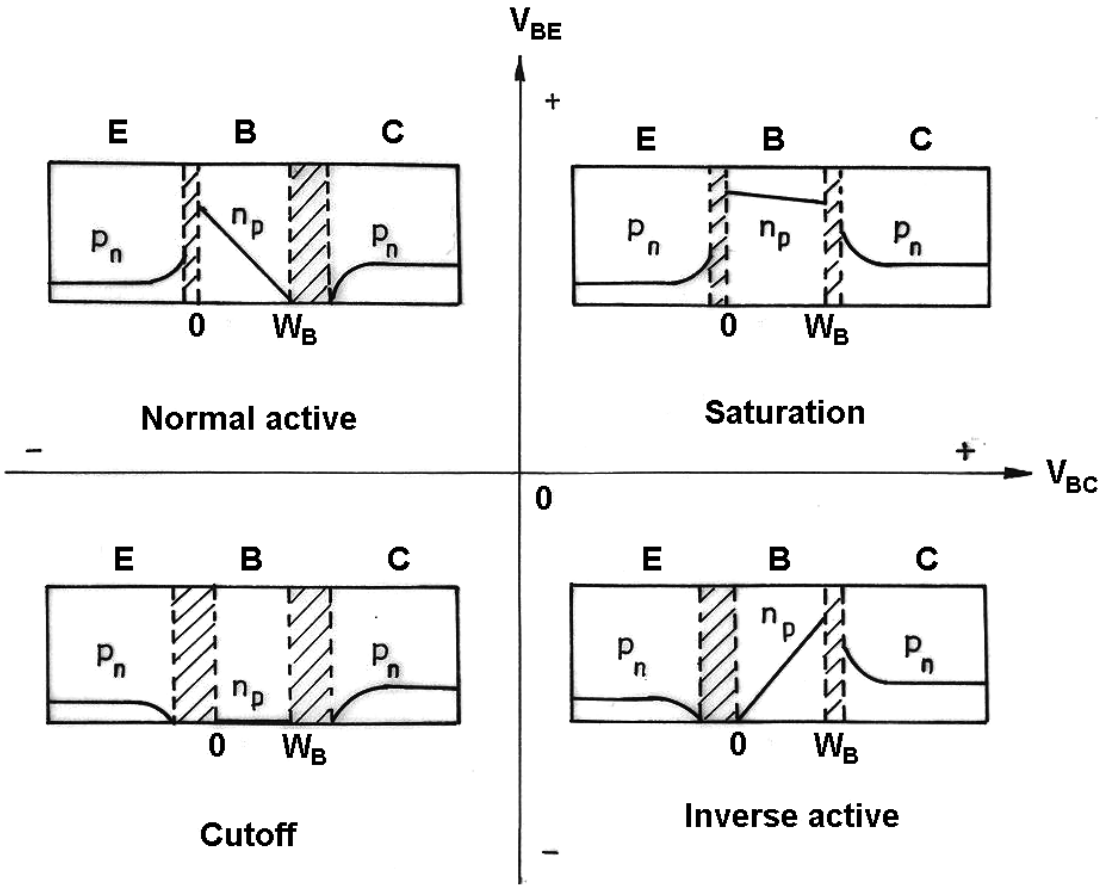


Fig. 6.4: Minority carrier concentration profiles in emitter, base and collector regions of an npn transistor for different polarities of V_{BE} and V_{BC}

Due to the concentration gradients of the minority carriers in the neutral regions, there will be flow of diffusion currents. The minority carrier electron and hole current components at the E-B junction are denoted by I_{nE} and I_{pE} respectively. When the E-B junction is forward biased ($V_{BE} > 0$), holes are injected from the base to the emitter and I_{pE} is negative (i.e. in the negative x-direction). On the other hand, when $V_{BE} < 0$, the hole concentration in the emitter reduces below the thermal equilibrium value as the minority carrier holes are swept to the base, giving rise to a positive I_{pE}. When $V_{BE} > V_{BC}$, due to the concentration gradient of electrons in the base, electrons will be injected from the

emitter to the base, giving rise to a negative I_{nE} (Note that the direction of current is opposite to the direction of electron flow). However, when $V_{BE} < V_{BC}$, the direction of the concentration gradient changes and electrons may be injected from base to emitter resulting in a positive I_{nE}. (There is a special case when $V_{BE} \approx V_{BC}$, which we shall discuss later in Section 6.6.2). For the C-B junction also, there exists electron and hole current components denoted by I_{nC} and I_{pC} respectively. Using similar arguments as for the E-B junction, I_{pC} will be positive when $V_{BC} > 0$, and will be negative when $V_{BC} < 0$. I_{nC} will in general be negative when $V_{BE} > V_{BC}$, and positive when $V_{BE} < V_{BC}$. (Note that positive or negative values of the current components only refer to their direction of flow. For positive hole current, the flow of holes is in the positive x-direction, while for positive electron current, the flow is in the negative x-direction.).

We have already seen in Section 6.2 that for proper transistor action, most of the electrons injected into the base at one of the junctions should reach the other junction. This implies that there is dependence between I_{nE} and I_{nC}. However, $I_{nE} \neq I_{nC}$. For example, when electrons injected from the E-B junction flow towards the C-B junction, they partly recombine with the majority carrier holes in the base, and therefore $I_{nE} \neq I_{nC}$. The difference, $I_{BR} = -(I_{nE} - I_{nC})$ is referred to as the base recombination current. An equivalent hole current flows in from the base terminal to compensate for the holes lost during the base recombination process. Therefore, the base current consists of the following components : (a) hole current at the E-B junction (I_{pE}); (b) hole current at the C-B junction (I_{pC}) and (c) base recombination current (I_{BR}). The emitter current is due to (a) the hole current at the E-B junction (I_{pE}) and (b) the electron current at the E-B junction (I_{nE}), while the collector current is due to (a) the hole current at the C-B junction

(I_{pC}) and (b) the electron current at the C-B junction (I_{nC}). Hence, considering that I_E and I_C actually flow in the negative x-direction (refer to Fig.6.3), we can write

$$\left.\begin{aligned} I_E &= -(I_{nE} + I_{pE}) \\ I_B &= I_{pC} - I_{pE} + I_{BR} = I_{pC} - I_{pE} - (I_{nE} - I_{nC}) \\ I_C &= -(I_{nC} + I_{pC}) \end{aligned}\right\} \quad \text{.....(6.1)}$$

Rearranging, from eqn.(6.1) we get

$$I_E = I_B + I_C \text{ ……..(6.2)}$$

Eqn. (6.2) states that the total current flowing into the transistor is equal to the current flowing out, and is only the Kirchoff's Current Law as applied to BJTs. Fig. 6.5 shows the various current components and the direction of flow of carriers in the normal active mode of operation, i.e. when $V_{BE} > 0$ and $V_{BC} < 0$.

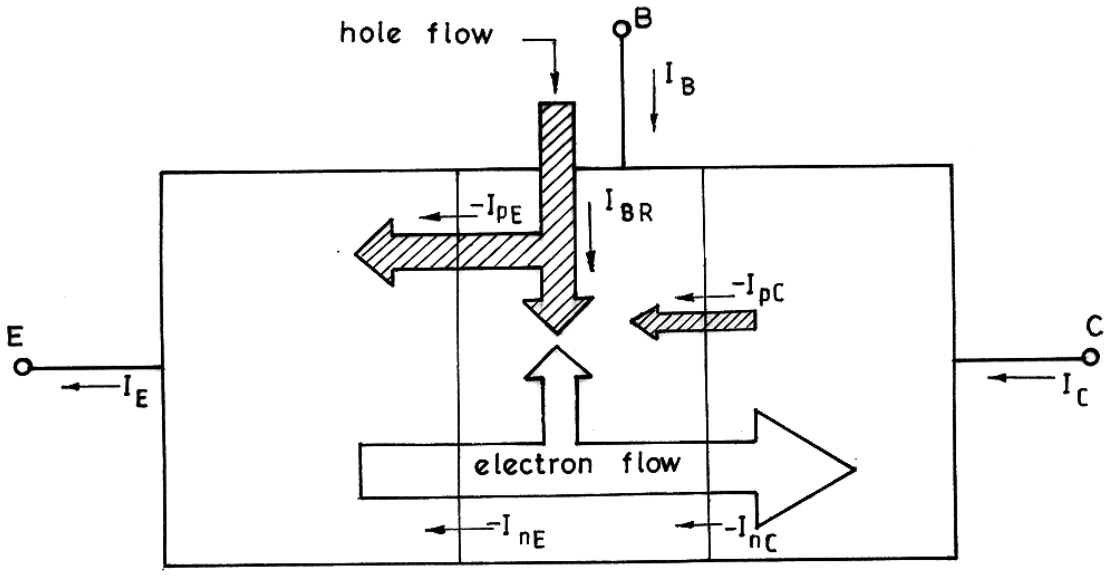


Fig. 6.5: Current components in an npn transistor in the normal active mode of operation.

Q6.1 Why have we neglected the majority carrier current components in eqn. (6.1) while developing expressions for the terminal currents.

We **have not** neglected the majority carrier current components. By considering only the minority carrier components at the junctions, it is possible to obtain the total current at the contacts. We are therefore able to avoid solving the continuity equation for the majority carriers, which is very difficult to handle. We have also used a similar procedure for diodes. As an example, let us consider the emitter current in a BJT, which is composed of a majority carrier (electron) current ($I_{n\text{-}con}$) and a minority carrier (hole) current ($I_{p\text{-}con}$) at the emitter contact. The electrons injected at the emitter contact, while flowing towards the E-B junction, partly recombine in the bulk emitter. Since I_{nE} is the electron current at the E-B junction, which is injected as a minority carrier current into the base, we can write $I_{nE} = I_{n\text{-}con} - I_{BE}$, where I_{BE} is the recombination current in the bulk emitter. Similarly, the holes injected into the emitter partly recombine, while flowing towards the emitter contact. Since an identical number of holes and electrons are involved in recombination, we can write $I_{p\text{-}con} = I_{pE} - I_{BE}$, where I_{pE} is the hole current at the junction. Therefore, we see that $I_{n\text{-}con} + I_{p\text{-}con} = I_{nE} + I_{pE}$, or **the sum of the majority and minority carrier components at the contact equals the sum of the minority carrier current components at the junction.**

6.5 Basic BJT Parameters

A bipolar transistor can be operated in any of the three configurations shown in fig. 6.7 (a), (b) and (c). In Fig.6.7(a), the base terminal is common to both the input and

the output and is hence termed the common-base configuration. Similarly, the configurations in Figs.6.7(b) and (c), where the emitter and collector are the common terminals, are called common-emitter and common-collector configurations respectively. Fig.6.7 also shows the input and output currents and voltages for each configuration.

One of the most important parameters of the BJT is its **common-base current gain (α)** defined for the normal active region of operation as

$$\alpha = \left| \frac{I_{nC}}{I_E} \right| \ldots.(6.30)$$

From eqn. (6.1), we can therefore express I_C as

$$I_C = -(I_{nC} + I_{pC}) = \alpha I_E - I_{pC} = \alpha I_E + I_{CBO} \ldots.(6.31)$$

where I_{CBO} is the small leakage current flowing in the reverse biased C-B junction when $I_E = 0$, i.e. the emitter contact is open-circuited. The first two subscripts (CB) in I_{CBO} refer to the two terminals between which the current is flowing and the third subscript (O) refers to the open-circuited third terminal. Neglecting the small leakage current I_{CBO}, we see that α is the ratio of the output current (I_C) to the input current (I_E) in the common base configuration, and is therefore referred to as the common base current gain.

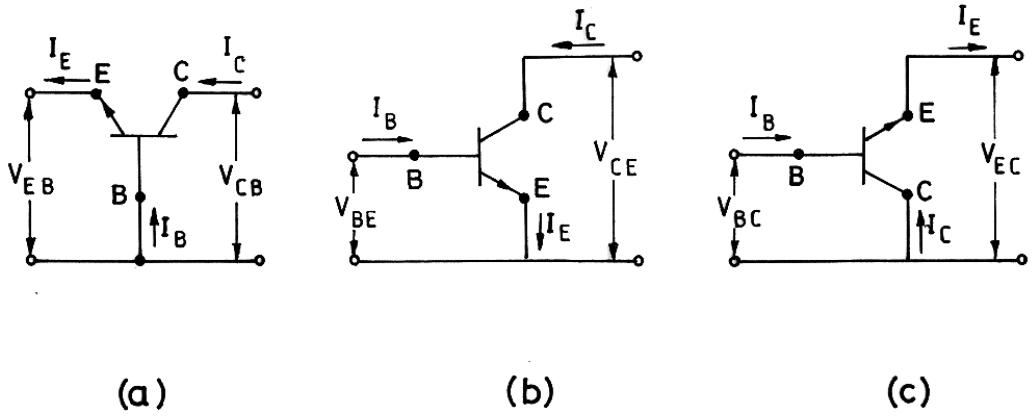


Fig. 6.7: An npn transistor in (a) common-base, (b) common-emitter and (c) common-collector configuration

The parameter α is dependent on two factors, viz. the **emitter efficiency (γ)**, and the **base transport factor (α_T)**. The emitter efficiency (γ) is defined as the ratio of the current due to electrons injected from the emitter into the base (I_{nE}) to the total emitter current (I_E), i.e.

$$\gamma \equiv \left|\frac{I_{nE}}{I_E}\right| = \frac{1}{1+\frac{I_{pE}}{I_{nE}}} \quad \text{.....(6.32)}$$

An emitter efficiency close to unity indicates that almost the entire emitter current is due to the "useful" injection of electrons from the emitter to the base. The Base transport factor (α_T) is defined as the ratio of the collector current due to injected electrons from the emitter (I_{nC}) to the current due to electrons injected from emitter into base (I_{nE}), i.e.

$$\alpha_T \equiv \left|\frac{I_{nC}}{I_{nE}}\right| = \frac{1}{1+\left|\frac{I_{BR}}{I_{nC}}\right|} \quad \text{.....(6.33)}$$

If almost all the electrons injected into the base reaches the collector junction without recombining, α_T becomes close to unity. From eqns. (6.30), (6.32) and (6.33) we see that

$$\alpha = \gamma\alpha_T \ldots\ldots..(6.34)$$

It may be noted that ideally the maximum values of α_T and γ are 1. However, in all practical situations, they are less than (but very close to) 1. Thus the value of α is also close to unity in most well-designed BJTs.

The other important parameter is the **common-emitter current gain (usually represented as β or h_{FE})** defined as

$$\beta = \frac{\alpha}{1-\alpha} \ldots\ldots(6.35)$$

As α is close to unity, β of transistors can be quite large and often in the range of 100-500. Using eqns.(6.2), (6.31) and (6.35), it can be shown that

$$I_C = \beta I_B + (\beta+1)I_{CBO} = \beta I_B + I_{CEO} \ldots\ldots(6.36)$$

where $I_{CEO} = (\beta+1)I_{CBO}$ is the leakage current flowing between collector and emitter terminals with the base terminal open circuited. Neglecting the leakage current I_{CEO}, we see that β is the ratio of the output current (I_C) to the input current (I_B) in the common emitter configuration, and is therefore referred to as the common emitter current gain.

Q.6.4: Show that α can be expressed as $\frac{\beta}{1+\beta}$

From eqn. (6.7), we can write $\beta(1-\alpha) = \alpha$.

Rearranging, we get $\alpha(1+\beta) = \beta$ or $\alpha = \frac{\beta}{1+\beta}$

For a practical transistor, assuming both the base and emitter widths to be much smaller than the respective minority carrier diffusion lengths, we get

$$\frac{I_{pE}}{I_{nE}} \approx \frac{D_{pE} N_A W_B}{D_n N_{DE} W_E} \quad \text{.....(6.38)}$$

The emitter efficiency (γ) can now be obtained by substituting eqn (6.38) into eqn. (6.32) as

$$\gamma = \frac{1}{1 + \frac{D_{pE} N_A W_B}{D_n N_{DE} W_E}} = \frac{1}{1 + \frac{D_{pE}(GN_B)}{D_n (GN_E)}} \quad \text{.....(6.39)}$$

From the above equation we observe that the maximum value of γ is 1. In order to obtain γ close to 1, the doping concentration in the base must be made much smaller than that in the emitter (i.e. $N_A << N_{DE}$). Also, the base width must be much smaller than the emitter width (i.e. $W_B << W_E$). In fact, the number of dopants per unit area, called the **Gummel number** (GN), is an important design parameter. For emitter efficiency close to unity, the emitter Gummel number ($GN_E = N_{DE}W_E$) must be much greater than the base Gummel number ($GN_B = N_A W_B$).

The base transport factor α_T can be obtained by substituting the expressions for I_{nC} and I_{BR} using the condition $W_B << L_n$. The base transport factor can then be approximated as

$$\alpha_T \approx \frac{1}{1 + \frac{W_B^2}{2L_n^2}} \quad \text{.....(6.41)}$$

From eqn. (6.41), we see that in order to have $\alpha_T \approx 1$, W_B must be much less than L_n so that recombination in the base is minimized. Now, substituting eqns. (6.39) and (6.41) in eqn.(6.34) and making suitable approximations, α can be expressed as

$$\alpha = \frac{1}{\left(1+\frac{D_{pE}N_A W_B}{D_n N_{DE} W_E}\right)\left(1+\frac{W_B^2}{2L_n^2}\right)} \approx \frac{1}{1+\frac{D_{pE}N_A W_B}{D_n N_{DE} W_E}+\frac{W_B^2}{2L_n^2}} \quad \text{……..(6.42)}$$

The common-emitter current gain (β) can now be expressed by substituting eqn. (6.42) into eqn.(6.35) . Hence we have

$$\beta = \frac{\alpha}{1-\alpha} = \frac{1}{\frac{1}{\alpha}-1} = \frac{1}{\frac{D_{pE}N_A W_B}{D_n N_{DE} W_E}+\frac{W_B^2}{2L_n^2}} \quad \text{…(6.43)}$$

In order to obtain a high value of β (and also α close to 1), the denominator in eqn. (6.43) must be made small. Therefore, the following conditions must be satisfied: (a) $N_A << N_{DE}$; (b) $W_B << W_E$; (c) $W_B << L_n$. The first two conditions ensure that the hole injection from the base into the emitter is negligible compared to the forward injection of electrons from the emitter into the base, resulting in $\gamma \approx 1$. Also the base should be very narrow so that almost all the electrons injected by the emitter can reach the collector without recombining in the base and we have $\alpha_T \approx 1$.

Now, considering a transistor with $N_A << N_{DE}$, from eqns. (6.6) and (6.12), we have $n'_{pE} >> p'_{nE}$. Also since the C-B junction is reverse biased in the normal active mode, $n'_{pE} >> p'_{nC}$. Therefore, we can simplify eqn.(6.28) by neglecting terms involving p'_{nE} and p'_{nC}, so that

$$I_B \approx I_{BR} = \frac{Aqn'_{pE}W_B}{2\tau_n} = \frac{Q_{nB}}{\tau_n} \quad \text{…..(6.44)}$$

where Q_{nB} is given by eqn.(6.24). Now, neglecting I_{CEO} in eqn.(6.36) and from eqns. (6.29) and (6.44), we have

$$\beta \approx \frac{I_C}{I_B} \approx \frac{\tau_n}{\tau_{tB}} \quad \ldots\ldots(6.45)$$

Thus we see that the expression for β can be extremely simplified under certain conditions to a ratio of two time constants, viz. the electron lifetime in the base and the base transit time.

We have seen in this Section that both α and β, **which are defined in the normal active mode of operation**, are constants depending only on the physical parameters of the BJT. This implies that when a BJT is operating in the normal active mode, its current gain is independent of the bias voltages.

Example 6.1 An n-p-n Bipolar Junction Transistor T1 has the following parameters:

	Emitter (n-type)	Base (p-type)	Collector (n-type)
Width (μm)	1.0	0.8	3.0
Doping concentration (/cm^3)	10^{19}	10^{17}	$5\text{x}10^{15}$
Minority carrier diffusion constant (cm^2/sec)	2	20	10
Minority carrier lifetime(μs)	0.01	0.1	1

Calculate the emitter efficiency (γ), base transport factor (α_T), common base current gain (α) and common emitter current gain (β) for this transistor.

The minority carrier diffusion lengths in the emitter, base and collector are given by

$L_{pE} = \sqrt{2\text{x}0.01\text{x}10^{-6}}\text{cm} = 1.4\mu\text{m}$, $L_{nB} = \sqrt{20\text{x}0.1\text{x}10^{-6}}\text{cm} = 14\mu\text{m}$ and

$L_{pC} = \sqrt{10\text{x}10^{-6}}\text{cm} = 31.6\mu\text{m}$.

Since W_E is comparable to L_{pE}, we cannot use eqn.(6.39) to evaluate γ. From eqn. (6.32) and eqn.(6.37), and since $W_B << L_n$, we can write

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$$\gamma = \frac{1}{1+\dfrac{D_{pE} W_B N_A \coth\left(\dfrac{W_E}{L_{pE}}\right)}{D_n L_{pE} N_{DE}}} = \frac{1}{1+\dfrac{2x0.8x10^{-4}\, x10^{17} \coth\left(\dfrac{1}{1.4}\right)}{20x1.4x10^{-4}\, x10^{19}}} = 0.999$$

Since $W_B << L_n$, from eqn.(6.41), we have

$$\alpha_T \approx \frac{1}{1+\dfrac{W_B^2}{2L_n^2}} = \frac{1}{1+\dfrac{\left(0.8x10^{-4}\right)^2}{2\left(14x10^{-4}\right)^2}} = 0.99837$$

From eqn. (6.34), $\alpha = \gamma\alpha_T = 0.999x0.99837 = 0.9974$

From eqn. (6.35), $\beta = \dfrac{\alpha}{1-\alpha} = \dfrac{0.9974}{1-0.9974} = 383.6$

6.6 The Ebers-Moll model

For circuit simulation, it is necessary to have a common set of equations, which relate the terminal currents to the terminal voltages, for all modes of operation. Also, the model should have few parameters which can be extracted experimentally. The Ebers-Moll model is ideally suited for this purpose and is included in the popular circuit simulator SPICE.

The basic equivalent circuit used for the Ebers-Moll model of the BJT is shown in Fig. 6.8. It consists of two back-to-back diodes. The currents through the diodes between the E-B and C-B terminals are denoted by I_F and I_R respectively and are expressed by the usual diode equation, i.e.

$$I_F = I_{ES}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \ldots\ldots (6.46)$$

$$I_R = I_{CS}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \ldots\ldots (6.47)$$

where I_{ES} and I_{CS} are the reverse saturation currents of the E-B and B-C junction diodes respectively. In addition, there are two current sources which stand for the coupling between the two junctions. The current source between the C-B terminals is related to the diode current at the E-B junction through the **forward common base current gain (α_F)**. Similarly, the current source between the E-B terminals is related to the diode current at the C-B junction through the **reverse common base current gain (α_R)**. For a symmetrical transistor having identical emitter and collector junction area, doping and other parameters, α_F and α_R will be identical. This means that the transistor will behave in exactly the same way if the biasing of the emitter and the collector junction is interchanged. However, practical transistors are not symmetric and hence α_F and α_R have different values.

Now, from Fig. 6.8, the emitter, collector and base currents in a BJT can be modeled as

$$I_E = I_F - \alpha_R I_R \ldots\ldots(6.48)$$

$$I_C = \alpha_F I_F - I_R \ldots\ldots.(6.49)$$

$$I_B = I_E - I_C = (1-\alpha_F)I_F + (1-\alpha_R)I_R \ldots.(6.50)$$

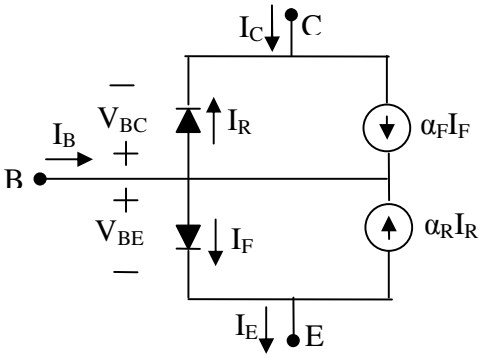


Fig. 6.8: Basic Ebers –Moll equivalent circuit of a BJT

In a modern BJT, all the three regions (i.e. emitter, base and collector) have very small widths. Assuming that their widths are much smaller than the respective diffusion lengths, the minority carrier distributions can be approximated by linear profiles as shown in Fig.6.9 (refer to Q. 6.2). In this figure, Q_E, Q_{BF}, Q_{BR} and Q_C refer to the stored charge in the emitter due to holes injected from the base, the stored charge in the base due to electrons injected from the emitter, the stored charge in the base due to electrons injected from the collector and the stored charge in the collector due to holes injected from the base respectively. Note that the total stored charge in the base is the sum of Q_{BF} and Q_{BR}. With the help of the equivalent circuit and the minority carrier distribution profile shown in Figs.6.8 and 6.9, it is now possible to model the currents in a BJT for all possible combinations of V_{BE} and V_{BC}. We shall first analyze the two limiting cases when one junction is short-circuited and a bias is applied only across the other junction, and then the general case of non-zero voltages applied to both the junctions.

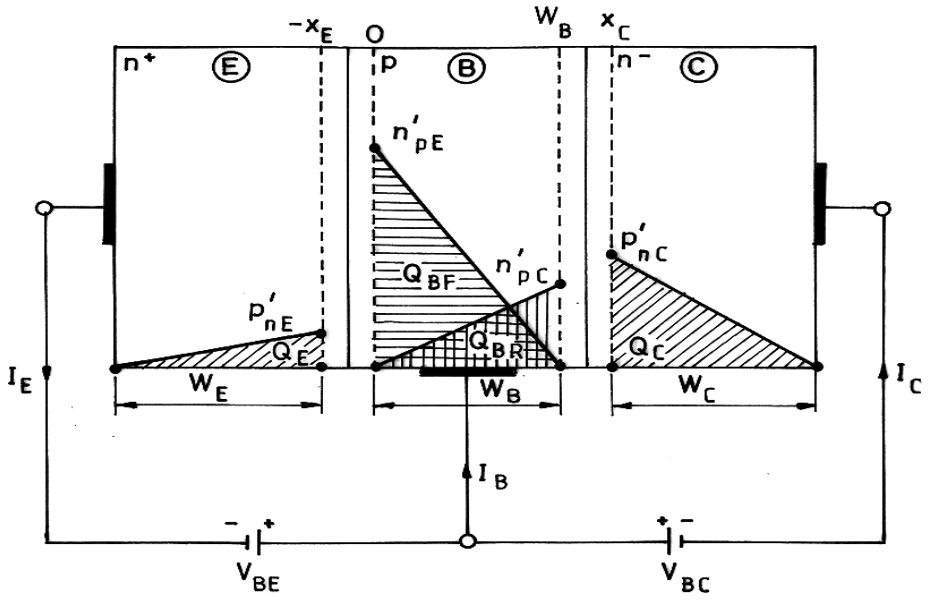


Fig. 6.9: Stored minority carrier charges in the emitter, base and collector of a BJT

Case 1: $V_{BE} \neq 0$, $V_{BC} = 0$

This is called the forward operation of the transistor. For $V_{BC} = 0$, from eqn. (6.47), $I_R = 0$. Therefore from eqns. (6.48) to (6.50), we get

$$I_C = \alpha_F I_F \ldots(6.51)$$

$$I_E = I_F \ldots..(6.52)$$

$$I_B = I_E - I_C = (1-\alpha_F) I_F \ldots.(6.53)$$

The **forward common emitter current gain (β_F)**, defined as the ratio of collector current to base current when $V_{BC} = 0$, is given by

$$\beta_F = \frac{I_C}{I_B} = \frac{\alpha_F}{1-\alpha_F} \ldots..(6.54)$$

Also, we note that for $V_{BC} = 0$, $Q_{BR} = Q_C = 0$. Under these circumstances, the collector current can be expressed as

$$I_C = \alpha_F I_F = AqD_n \frac{dn}{dx} = AqD_n \frac{n'_{pE}}{W_B} = \frac{AqD_n n_i^2}{N_A W_B}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] (6.55)$$

where

$$I_S = \frac{AqD_n n_i^2}{N_A W_B} \ldots\ldots(6.56)$$

Eqn.(6.55) is essentially same as eqn. (6.29). Now the base current for this case is the sum of the recombination current I_{BR} and the injected hole current into emitter I_{pE}. I_{pE} can be expressed as

$$I_{pE} = AqD_p \frac{dp}{dx} = \frac{AqD_p n_i^2}{N_{DE} W_E}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \ldots\ldots(6.57)$$

Noting that $Q_{nB} = Q_{BF}$ for this case, using $I_{BR} = \frac{Q_{nB}}{\tau_n}$ as given by eqn. (6.25), we have

$$I_B = I_{pE} + I_{BR} = \frac{AqD_p n_i^2}{N_{DE} W_E}\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right] + \frac{Aqn_i^2 W_B}{2N_A \tau_n}\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right] \quad \ldots\ldots(6.58)$$
$$= Aqn_i^2\left(\frac{D_p}{N_{DE} W_E} + \frac{W_B}{2N_A \tau_n}\right)\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]$$

Using eqns. (6.54), (6.55) and (6.58), we obtain the expression for β_F, which is identical to the expression for the common emitter gain β given by eqn. (6.43).

Case2: $V_{BE} = 0$, $V_{BC} \neq 0$

This is the reverse operation of the transistor. For $V_{BE} = 0$, from eqn.(6.46), $I_F = 0$. Also, $Q_E = Q_{BF} = 0$. The stored charge in the base is represented by Q_{BR} and the emitter current is due to electrons injected into the base from the collector. Therefore,

$$I_E = -\alpha_R I_R = -AqD_n \frac{dn}{dx} = -\frac{AqD_n n_i^2}{N_A W_B}\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right] = -I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right] \ldots.(6.59)$$

We note that the direction of the emitter current has changed. This is because electrons are now injected from the collector into the base and flows towards the emitter due to the concentration gradient in the base. It is now possible to define a parameter called the **reverse common emitter current gain (β_R)**, which is essentially the ratio of the emitter current to the base current when $V_{BE} = 0$ and is given by $\beta_R = \frac{\alpha_R}{1-\alpha_R}$.

Case3: $V_{BC} \neq 0$, $V_{BE} \neq 0$

In order to analyze this general condition we use superposition of the currents obtained under the two limiting cases discussed above. Thus using eqns. (6.49), (6.55) and (6.59), we can write

$$I_C = \alpha_F I_F - I_R = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right] - \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right]}{\alpha_R} \ldots\ldots(6.60)$$

From the expression of β_R, we see that $\alpha_R = \frac{\beta_R}{1+\beta_R}$. Thus eqn. (6.60) can be written as

$$I_C = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right)\right] - \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right]}{\beta_R} \ldots.(6.61)$$

Similarly the emitter current can be expressed from eqns. (6.48), (6.55) and (6.59) as

$$I_E = I_F - \alpha_R I_R = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]}{\alpha_F} - I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right] \ldots..(6.62)$$

Considering $\alpha_F = \frac{\beta_F}{1+\beta_F}$, we can rewrite eqn.(6.62) as

$$I_E = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right)\right] + \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]}{\beta_F} \ldots\ldots.(6.63)$$

Therefore the expression for the base current is obtained from $I_B = I_E - I_C$ as

$$I_B = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]}{\beta_F} + \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right]}{\beta_R} \ldots.(6.64)$$

From the symmetry of the expressions for the collector current and the emitter current, it can be seen that the equivalent circuit of a BJT can now be represented by two coupled diodes and only one current source as shown in Fig.6.10. According to this circuit, the currents flowing through the back-to-back diodes are I_{EC}/β_R and I_{CC}/β_F. The current

supplied by the current source is I_{CT}. From this circuit, the collector, emitter and base currents of a BJT can be expressed as

$$I_C = I_{CT} - \frac{I_{EC}}{\beta_R} \ldots\ldots (6.65)$$

$$I_E = I_{CT} + \frac{I_{CC}}{\beta_F} \ldots\ldots (6.66)$$

$$I_B = \frac{I_{CC}}{\beta_F} + \frac{I_{EC}}{\beta_R} \ldots\ldots (6.67)$$

Referring to equations (6.61), (6.63) and (6.64) we can see that

$$I_{CT} = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right)\right], I_{CC} = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \text{ and } I_{EC} = I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right].$$

Thus we see that from the Ebers-Moll model we can obtain the values of the terminal currents for any combination of V_{BE} and V_{BC}, provided we know the values of **only three parameters, viz. I_S, β_F and β_R.**

The values of these parameters can be extracted very easily through simple experiments. If we short-circuit the collector-base junction (i.e. $V_{BC} = 0$) and vary V_{BE}, for $V_{BE} > 3V_T$, we see from eqns. (6.61) and (6.64) that I_C and I_B will vary as $I_C \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right)$ and $I_B \approx \frac{I_S}{\beta_F}\exp\left(\frac{V_{BE}}{V_T}\right)$. Now, if we plot $\ln(I_C)$ and $\ln(I_B)$ vs. V_{BE}, we shall get two parallel straight lines, as shown in Fig. 6.11. (Note that the lines will be parallel only in a small range of V_{BE}. The reason for this shall be discussed in Chapter 7.) The intercept of the $\ln(I_C)$ vs. V_{BE} curve, when extrapolated, gives $\ln(I_S)$. The ratio of I_C and I_B gives β_F. A similar experiment where $V_{BE} = 0$ and V_{BC} is varied can be used to extract β_R.

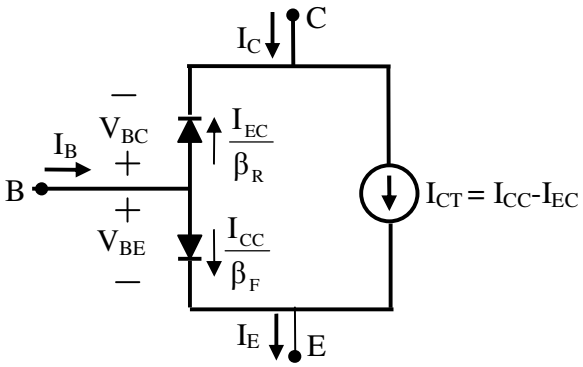


Fig. 6.10: Three-parameter Ebers-Moll equivalent circuit of a BJT

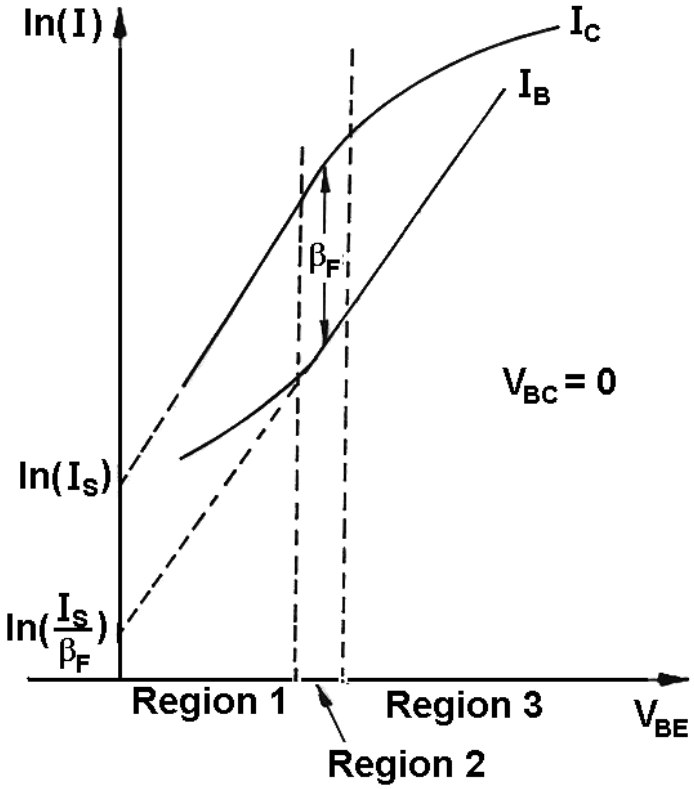


Fig. 6.11 $\ln(I_C)$ and $\ln(I_B)$ versus V_{BE} characteristics of a BJT when $V_{BC} = 0$

Q 6.5 The E-B junction of an n-p-n transistor is forward biased by V_{BE}, while the collector terminal is left floating. (a)What is the open circuit voltage developed across the C-B junction? (b)What is the value of V_{CE} at this point? (c)What is the mode of operation of this BJT?

(a) From eqn. (6.65), when $I_C = 0$, $I_{CT} = I_{EC}/\beta_R$. Substituting the expressions for I_{CT} and I_{EC}, we have $\left(1+\frac{1}{\beta_R}\right)\exp\left(\frac{V_{BC}}{V_T}\right) = \exp\left(\frac{V_{BE}}{V_T}\right) + \frac{1}{\beta_R}$

Therefore

$$V_{BC} = V_T \ln\left[\frac{\beta_R \exp\left(\frac{V_{BE}}{V_T}\right)+1}{\beta_R+1}\right] = V_T \ln\left[\alpha_R\left\{\exp\left(\frac{V_{BE}}{V_T}\right)-1\right\}+1\right] \approx V_T \ln\left[\alpha_R \exp\left(\frac{V_{BE}}{V_T}\right)\right]$$

(b) From the above equation

$$\exp\left(\frac{V_{BC}}{V_T}\right) = \alpha_R \exp\left(\frac{V_{BE}}{V_T}\right), \text{ or } \exp\left(\frac{V_{BE}-V_{BC}}{V_T}\right) = \exp\left(\frac{V_{CE}}{V_T}\right) = \frac{1}{\alpha_R}$$

Therefore, $V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right)$

(c) We see that V_{BC} is positive, i.e. B-C junction is forward biased. Since both the junctions are forward biased, the transistor is in the saturation mode of operation.

Example 6.2 Calculate (a) V_{BC} and (b) V_{CE} when the collector terminal of transistor T1 (see Example 6.1) is kept floating and $V_{BE} = 0.6$V.

The expression for V_{BC} when $I_C = 0$ has been derived in Q 6.5. To calculate the open circuit V_{BC}, we need to know the value of α_R. Since the roles of emitter and collector are

interchanged in the reverse mode of operation, we can use eqn.(6.42) to calculate α_R by replacing the parameters of the emitter with that of the collector. Therefore

$$\alpha_R = \frac{1}{1+\frac{D_{pC}N_A W_B}{D_n N_{DC} W_C}+\frac{W_B^2}{2L_n^2}} = \frac{1}{1+\frac{10x10^{17}x0.8}{2x5x10^{15}x3}+\frac{0.8^2}{2x14^2}} = 0.2726$$

From Q. 6.5, $V_{BC} \approx V_T \ln\left[\alpha_R \exp\left(\frac{V_{BE}}{V_T}\right)\right] = 0.026\ln\left[0.2726\exp\left(\frac{0.6}{0.026}\right)\right] = 0.566\text{V}$

Also from Q 6.5, $V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right) = 0.026\ln\left(\frac{1}{0.2726}\right) = 0.034\text{V}$

Thus we see that when the collector is floating, V_{BC} is only slightly less than V_{BE}.

6.7 Static output I-V characteristics

The output characteristics of a BJT relate the output current to the output voltage for a fixed value of an input variable, usually the input current. Thus, for the common-emitter configuration, the output characteristics relate I_C to V_{CE} for a constant I_B. In this section we shall discuss the output characteristics of an n-p-n transistor in the common-base and common-emitter configurations.

6.7.1 Common-base configuration

As shown in Fig.6.12(a), the output characteristics of a BJT in the common base configuration plots the variation of the output current I_C versus the output voltage V_{CB} for a constant input current I_E. We shall now explain the nature of the characteristics from our understanding of the operation of the BJT, as discussed in the previous sections.

We know from Section 6.3 that I_E consists of two components, I_{pE} and I_{nE}. In a well designed transistor with emitter efficiency nearly equal to unity,

$I_E \approx -I_{nE} = \left(-AqD_n \frac{dn_p}{dx} \right)_{x=0}$ (refer to eqn.(6.8)). Thus, if I_E is fixed, $\left.\frac{dn_p}{dx}\right|_{x=0}$ will also remain at a constant value. Fig. 6.12(b) shows the minority carrier profiles in the base of a BJT for various values of V_{CB} while maintaining I_E as constant. We can see that $\left.\frac{dn_p}{dx}\right|_{x=0}$ is constant in all the plots.

Now, from eqn. (6.1), $I_C = -(I_{nC}+I_{pC})$. Let us first consider I_{nC}. We know that $-I_{nC} = \left(-AqD_n \frac{dn_p}{dx} \right)_{x=W_B} = -I_{nE} - I_{BR} \approx I_E - \frac{Q_{nB}}{\tau_n}$. Since I_E is held constant, this means that the rate of decrease in electron concentration $\left(-\frac{dn_p}{dx} \right)$ at x = W_B reduces with increase in stored charge in the base (Q_{nB}), which is actually the area under the minority concentration profile in Fig. 6.12(b). Also, it is clear that $-I_{nC}$, which is an important component of I_C, reduces with increase in Q_{nB}. In fact, when $\frac{Q_{nB}}{\tau_n} = I_E, \left.\frac{dn}{dx}\right|_{x=W_B} = 0$ and $I_{nC} = 0$. With this background, we can now explain the output characteristic curve and the plots of the minority carrier concentration in the base as shown in Fig. 6.12, where V_{CB} is varied in steps while keeping I_E constant.

Step1: $V_{CB} = 0$. Hence $n'_{pC} = 0$. Assuming recombination in the base to be small, the electron concentration in the base varies linearly as shown in plot1 in Fig.6.12(b). Since $I_{pC} = 0$, from eqn.(6.20), we can write $I_C = -I_{nC} = \frac{AqD_n n'_{pE}}{W_B}$. This value is shown as point 1 in the I_C vs. V_{CB} plot shown in fig. 6.12(a).

Step2: V_{CB} >> 0. A large reverse bias is applied across C-B junction. Now, $n'_{pC} \approx -n_{p0}$ (plot2 in Fig.6.12(b)). We can again approximate the electron concentration in the base by a linear variation, and at this point, $I_C = \frac{AqD_n\left(n'_{pE} + n_{p0}\right)}{W_B}$, as shown by point 2 in fig. 6.12(a). I_C cannot increase any further with further increase in the magnitude of V_{CB}. As $n_{p0} << n'_{pE}$, the difference in I_C between points 1 and 2 is very small.

Step3: $V_{CB} < 0$. A forward bias is applied across C-B junction. n'_{pC} is now positive and Q_{nB} increases. Therefore, as discussed before, the slope of the electron concentration profile at x = W_B reduces. Consequently $-I_{nC}$ and I_C decreases. In addition, due to an appreciable hole current (I_{pC}) from base to collector across the forward bias C-B junction, I_C is further reduced. The situation is depicted by point 3 and plot 3 in Figs. 6.12(a) and (b) respectively.

Step4: $V_{CB} = V_{CF}$, where V_{CF} is the value of V_{CB} when the collector terminal is floating. So, when $V_{CB} = V_{CF}$, $I_C = 0$ (point 4 in Fig. 6.12(a). We have already seen in Example 6.2 that $|V_{CF}| \approx V_{BE}$. Since Q_{nB} is increased further, there is a further reduction in the slope of the electron concentration profile at x = W_B as shown in plot 4 in Fig. 6.12(b). At this point, $I_{nC} = -I_{pC}$, so that $I_C = 0$.

Step5: $V_{CB} < 0$ and $|V_{CB}| > |V_{CF}|$. The minority carrier profile for this case is shown as plot 5 in Fig.6.12(b). As shown in the figure, the direction of the slope of the minority carrier concentration profile at x = W_B changes resulting in injection of electrons from collector to base. This, in addition to holes being injected from base to collector, results in a negative collector current as shown by point 5 in Fig. 6.12(a).

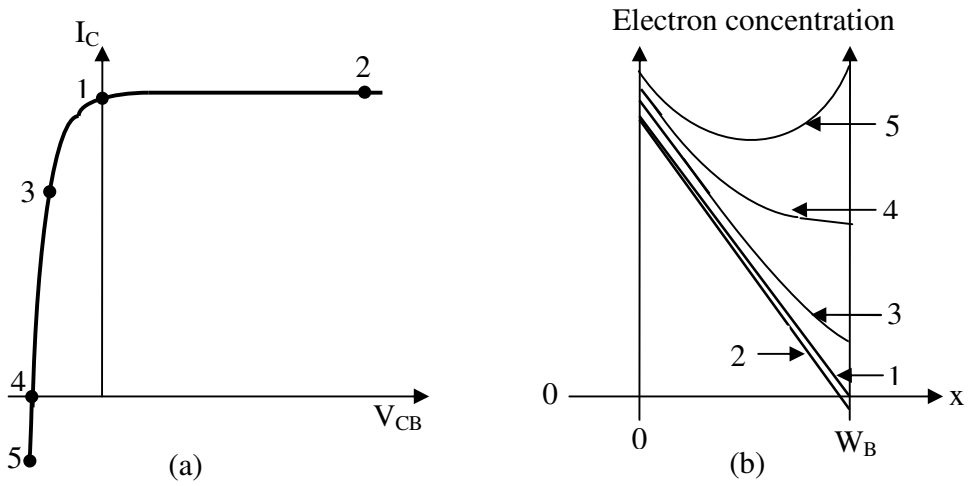


Fig.6.12: (a) Plot of I_C vs. V_{CB} and the corresponding (b) minority carrier concentration profile in the base of an npn transistor while keeping I_E constant.

We have now obtained the output characteristic curve for a particular I_E. Fig. 6.13 plots the output characteristic curves for various values of I_E. The figure also shows the different regions corresponding to the active, saturation and cut-off modes of operation. As seen from the figure, in the active region, there is a linear relationship between I_C and I_E as predicted by eqn.(6.31). Since $\alpha \approx 1$, we see that $I_C \approx I_E$ in the active region. In the saturation region, however, $I_C < \alpha I_E$.

6.7.2 Common-emitter configuration

As in the case of common-base mode, we shall now explain the output characteristics (I_C vs. V_{CE}) in the common-emitter configuration by varying the output voltage (V_{CE}) in a stepwise manner, while maintaining the input current (I_B) at a constant value. Fig.6.14(a) shows the output characteristics while Fig. 6.14(b) plots the minority carrier concentration profiles in the base for various values of V_{CE}, while keeping I_B constant.

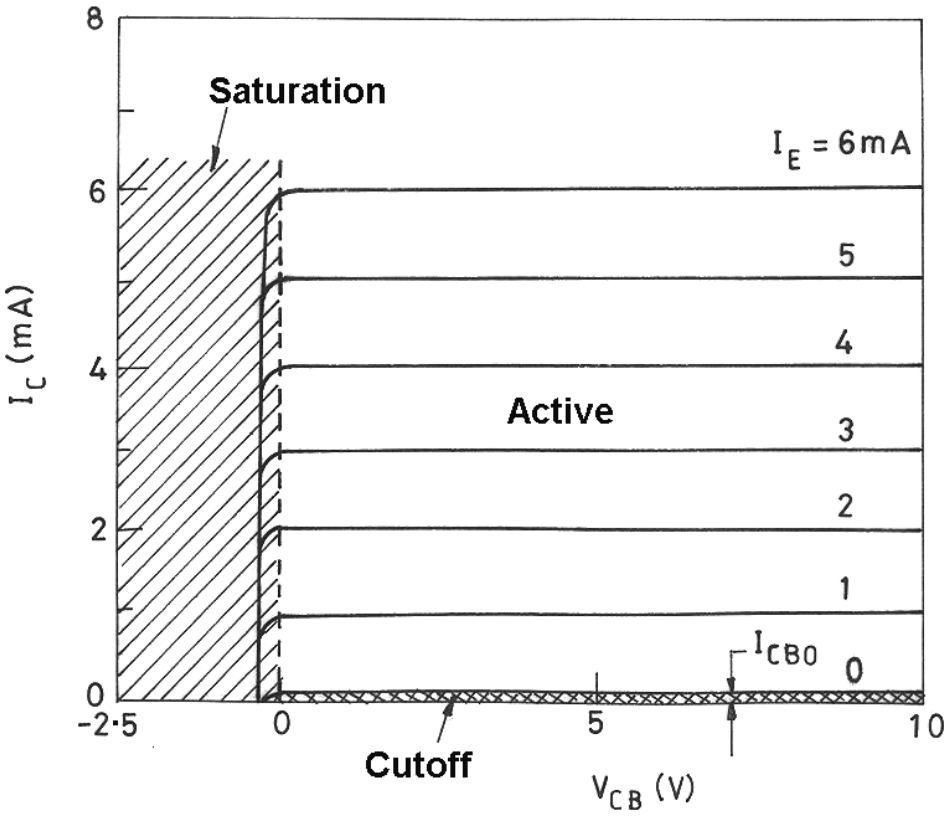


Fig. 6.13: Output characteristics (I_C vs V_{CB}) for npn transistor in common-base configuration for various values of I_E

We have already seen in Section 6.3 that I_B is composed of three components, viz. I_{pE}, I_{pC} and I_{BR}. We shall assume that the bipolar transistor has emitter efficiency almost equal to unity, and I_{pE} is negligibly small. Thus, when I_B is held constant, we can write

$$I_B \approx I_{BR} + I_{pC} = \frac{Q_{nB}}{\tau_n} + \frac{AqD_{pC}n_i^2 \coth\left(\frac{W_C}{L_{pC}}\right)}{L_{pC}N_{DC}}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] = \text{constant}$$

Therefore, if the C-B junction is reverse biased (i.e. $V_{BC} \leq 0$), the second term is almost constant and hence Q_{nB}, which is given by the area under the minority concentration

profile in Fig.6.14(b), is also almost a constant. On the other hand, increasingly forward biasing the C-B junction increases the value of the second term and hence results in a reduction in $Q_{nB.}$ This is reflected in the plots of Fig.6.14(b). We shall now explain the output characteristic curve and the minority carrier concentration plots as shown in Fig. 6.14, where V_{CE} is varied in steps while keeping I_B constant.

Step1: $V_{CE} = 0$, i.e. $V_{BC} = V_{BE}$. Now $n'_{pE} = n'_{pC}$. Although, it would seem that $I_{nC} = 0$, but this is not the case. This is because the variation in the electron concentration in the base is not linear, as shown in plot1 in Fig.6.14(b). The slope of the electron concentration at x = W_B implies that in addition to electrons being injected from the emitter, electrons are also injected from collector to base. Also, holes are injected from the base to the collector. Therefore I_C is negative as shown by point1 in Fig. 6.14(a). (Note that it is not possible to have a flat electron concentration profile in the base when $n'_{pE} = n'_{pC}$. If the profile were flat, it would mean that there is no electron injection either from emitter or collector. Since there is a large quantity of excess charge in the base, electrons are constantly recombining. With no supply of electrons, it is not possible to maintain the steady state condition of excess stored charges in the base).

Step2: $V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right)$. As already discussed in Q 6.5 and Example 6.2, the collector current is zero at this point (plot 2 in Fig. 6.14(b) and point 2 in Fig. 6.14(b)). This implies that the electron current is exactly equal and opposite to the hole current at the B-C junction, i.e. $|I_{nC}| - |I_{pC}| = 0$. The situation in the collector is similar to the example in Section 3.7.

Q. 6.6 When a transistor is in saturation and I_B is fixed, show that for small values of V_{CE} (<0.2V), V_{BE} increases and V_{BC} reduces with increase in V_{CE}. On the other hand, for larger values of V_{CE}, V_{BE} remains almost constant and the increase in V_{CE} is equal to the reduction in V_{BC}.

When I_B is fixed at a constant value (say I_{B1}), from eqn.(6.64), we have

$$I_{B1} = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]}{\beta_F} + \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right]}{\beta_R} \approx \frac{I_S}{\beta_F}\exp\left(\frac{V_{BE}}{V_T}\right)\left[1+\frac{\beta_F}{\beta_R}\exp\left(-\frac{V_{CE}}{V_T}\right)\right] ..(A)$$

In eqn. (A), as V_{CE} is increased, $\left[1+\frac{\beta_F}{\beta_R}\exp\left(-\frac{V_{CE}}{V_T}\right)\right]$ reduces, therefore $\exp\left(\frac{V_{BE}}{V_T}\right)$, and consequently V_{BE} has to increase to keep their product a constant. However, when V_{CE} > 0.2V, $\left[1+\frac{\beta_F}{\beta_R}\exp\left(-\frac{V_{CE}}{V_T}\right)\right] \approx 1$. Therefore from eqn. (A), V_{BE} becomes almost constant.

Eqn. (A) can also be rewritten as

$$I_{B1} = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]}{\beta_F} + \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right)-1\right]}{\beta_R} \approx \frac{I_S}{\beta_R}\exp\left(\frac{V_{BC}}{V_T}\right)\left[1+\frac{\beta_R}{\beta_F}\exp\left(\frac{V_{CE}}{V_T}\right)\right] ..(B)$$

In eqn. (B), $\left[1+\frac{\beta_R}{\beta_F}\exp\left(\frac{V_{CE}}{V_T}\right)\right]$ increases with increase in V_{CE}, therefore $\exp\left(\frac{V_{BC}}{V_T}\right)$ and consequently V_{BC} must reduce to keep their product constant. Since V_{BE} becomes constant for V_{CE} > 0.2V, from $V_{BC} = V_{BE} - V_{CE}$, any increase in V_{CE} results in an equal reduction in V_{BC}.

Step3: $\mathbf{V_T \ln\left(\frac{1}{\alpha_R}\right) < V_{CE} < V_{BE}}$. The situation is depicted by point 3 and plot 3 in Figs. 6.14(a) and (b) respectively. Since $V_{BC} = V_{BE} - V_{CE}$, and $V_{BE} > V_{CE}$, V_{BC} is positive, implying a forward biased B-C junction. Also, since $V_{BE} > V_{BC}$, $n'_{pE} > n'_{pC}$. The slope of the minority carrier concentration profile in the base depends on $n'_{pE} - n'_{pC}$. Now, $(n'_{pE} - n'_{pC}) \propto \exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right) = \exp\left(\frac{V_{BE}}{V_T}\right)\left[1 - \exp\left(-\frac{V_{CE}}{V_T}\right)\right]$. We have already seen in Q.6.6 that for small positive values of V_{CE}, V_{BE} increases with increasing V_{CE}. Thus, the slope of the electron concentration increases with increase in V_{CE}, resulting in an increased $|I_{nC}|$. From Q 6.6, we have also seen that the condition of constant I_B requires that in this range, V_{BC} is reduced with increasing V_{CE}, thereby reducing $|I_{pC}|$. As $I_C = |I_{nC}| - |I_{pC}|$, this would mean that I_C increases continuously with increase in V_{CE} in this range. However, the rate of increase reduces with increase in V_{CE}. As pointed out in Q.6.6, beyond $V_{CE} = 0.2V$, V_{BE} and consequently n'_{pE} is almost constant. Since $n'_{pC} << n'_{pE}$ in this range, the slope of the electron concentration and consequently $|I_{nC}|$ becomes almost constant. Also $|I_{pC}| << |I_{nC}|$, which implies that for $V_{CE} > 0.2V$, I_C is almost constant.

Step4: $\mathbf{V_{CE} = V_{BE}}$. Therefore, $V_{BC} = V_{BE} - V_{CE} = 0V$ and $n'_{pC} = 0$ as shown in plot 4 in Fig.6.14(b). As discussed previously, the electron concentration in the base now shows a linear dependence and since I_{pC} is negligibly small, the collector current now has a value of $I_C = \frac{AqD_n n'_{pE}}{W_B}$ shown as point 4 in fig. 6.14(a).

Step5: $V_{CE} > V_{BE}$. Now V_{BC} is negative, implying that the B-C junction is reverse biased. When V_{BC} is sufficiently negative, $n'_{pC} \approx -n_{p0}$ as shown in plot 5 in Fig.6.14(b). The collector current reaches a maximum value of $I_C = \frac{AqD_n\left(n'_{pE} + n_{p0}\right)}{W_B}$ (point 5 in Fig. 6.14(a)) and does not change any further with increase in V_{CE}. Since $n'_{pE} >> n_{p0}$, the increase in current from point 4 to point 5 in Fig.6.14(a) is very small. (In real transistors, however, there is an increase due to Early effect, which we shall discuss in Section 6.8).

As in the case of the common base configuration, the I_C-V_{CE} characteristics obtained for a fixed I_B can also be replicated for various values of I_B as shown in Fig. 6.15. As expected, I_C increases with increasing I_B. The figure also shows the saturation, active and cut-off regions of operation. The region where I_C is almost constant for a particular I_B is the active region. In this region, $I_C = \beta I_B$. In the saturation region, I_C reduces with reducing V_{CE} so that $I_C < \beta I_B$. The region of the characteristics where $I_C < I_{CEO}$ represents the cut-off mode of operation.

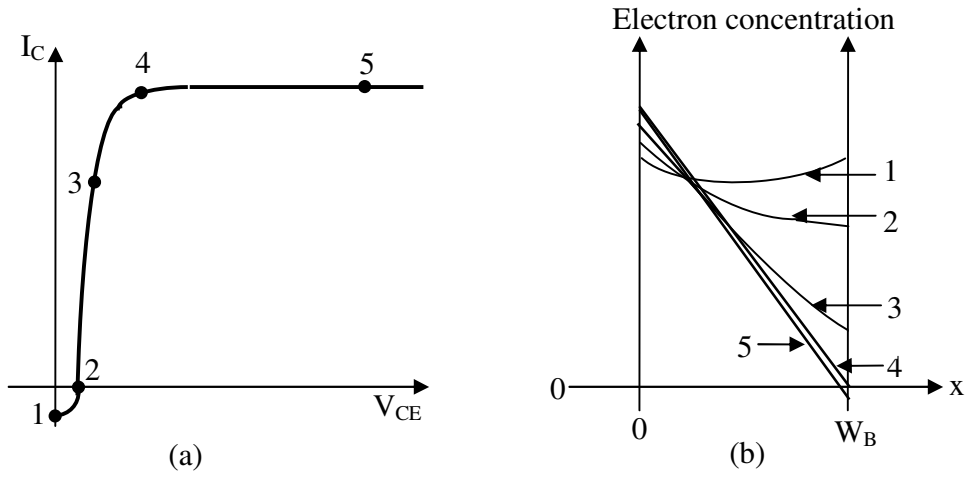


Fig.6.14: (a) Plot of I_C vs. V_{CE} and (b) the corresponding minority carrier concentration profile in the base of an npn transistor keeping I_B constant

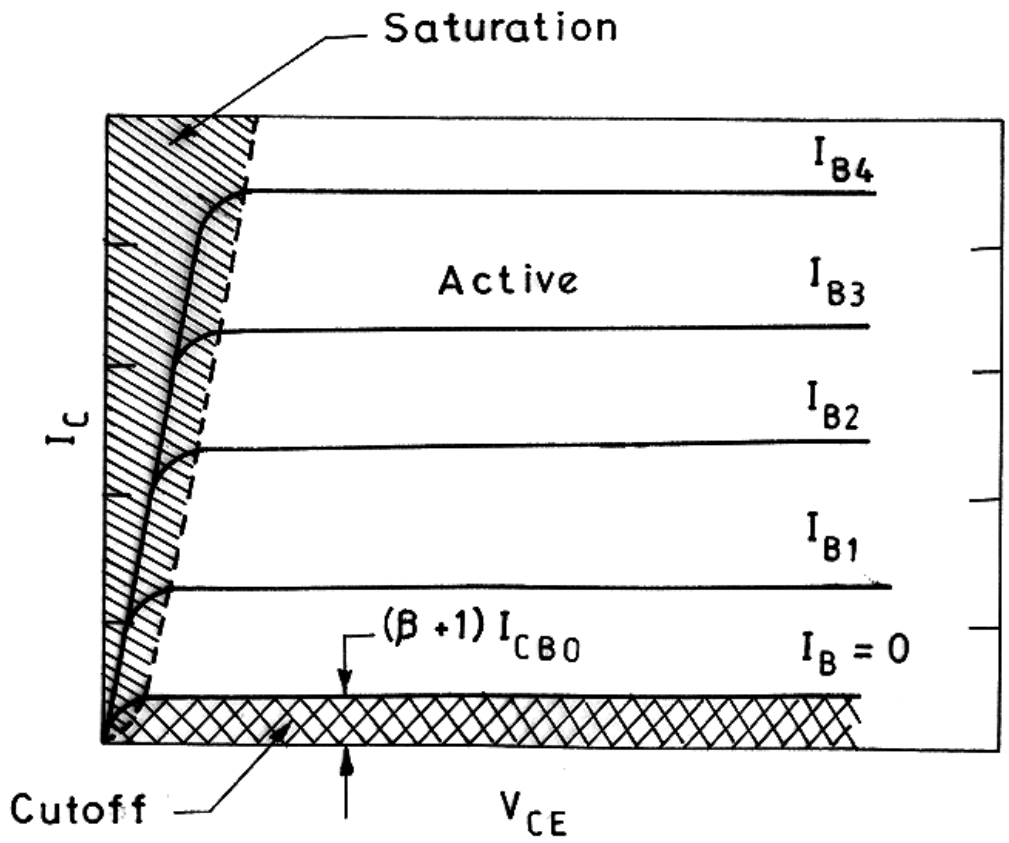
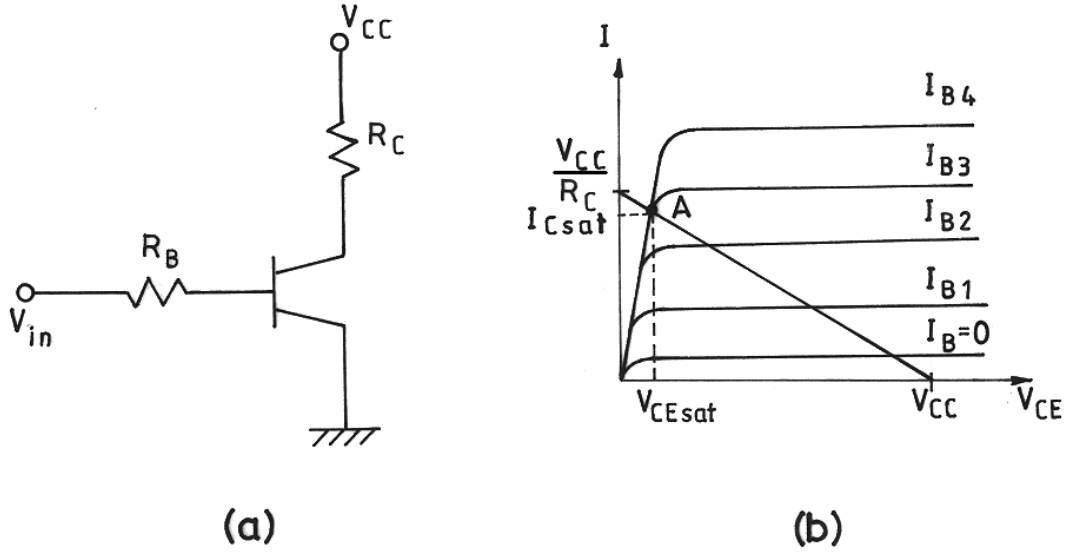


Fig. 6.15: Output characteristics (I_C vs V_{CE}) for npn transistor in common-emitter configuration for various values of I_B

Q 6.7 Why is the term 'saturation' used to denote the region where I_C is actually less than its maximum possible value for a particular I_B?

To explain the origin of this term, let us consider the circuit shown below, where the transistor has been connected in the common emitter configuration. In the given circuit, the input voltage is varied from 0V to a large positive value, resulting in a corresponding increase in I_B. In order to obtain the operating point (I_C, V_{CE}) corresponding to a particular I_B, we have also shown the output characteristics together with a load line. As usual, the operating point is at the intersection of the load line with the characteristic curve for the particular I_B. We see that when the transistor is operating

in the active region, $I_C = \beta I_B$, i.e. I_C increases linearly with I_B. However, with further increase in I_B, the operating point reaches the saturation region of the characteristics (point A), corresponding to $I_C = I_{Csat}$ and $V_{CE} = V_{CEsat}$. Now any further increase in I_B does not result in any increase in I_C, and I_C has saturated at its maximum value. This is how the saturation region of the characteristics got its name.



(a) (b)

6.8 Early Effect

For real transistors, there are some deviations from the characteristics predicted above by the simple analysis. In the common-base mode, for large reverse voltage applied across C-B junction, I_C was predicted to be independent of V_{CB}, i.e. $dI_C/dV_{CB} = 0$. In reality, I_C increases with an increase in V_{CB}. This is because as the reverse bias across C-B junction increases, the depletion layer width also increases. This effectively reduces the neutral base width from W_B to W_B'. Hence I_C increases due to an increase in the slope of the minority carrier concentration as shown in Fig.6.16. On the other hand, decrease in base width reduces I_B due to less recombination in the base. Consequently α increases

with increase in V_{CB}. The actual output characteristics of an n-p-n transistor in C-B mode is shown in Fig. 6.17(a) reflecting this increase in I_C. For the output characteristics in the common-emitter configuration, the effect is more dominant as the collector current is seen to rise significantly with an increase in V_{CE}. This is shown in Fig. 6.17(b).

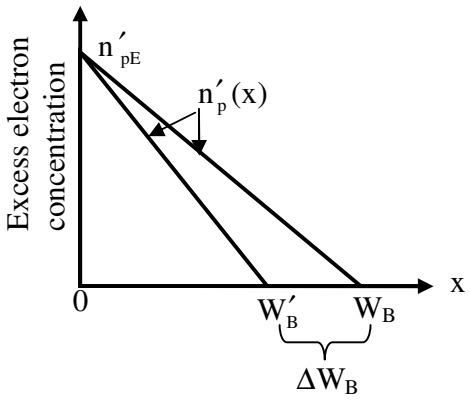


Fig. 6.16: Change in the slope of the minority carrier concentration profile due to modulation of neutral base width with change in V_{CB}

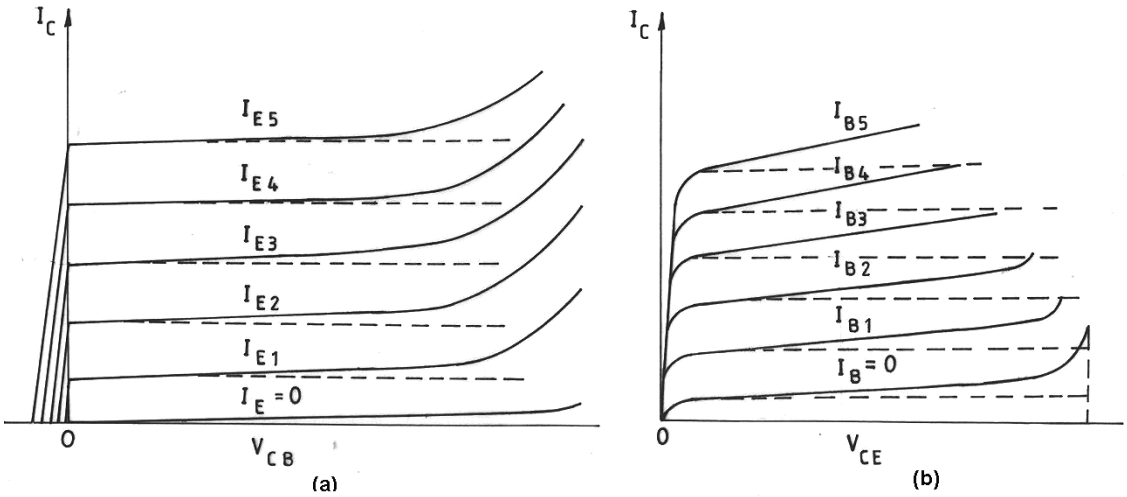


Fig. 6.17 : Practical output characteristics of an npn transistor in (a) common-base and (b) common emitter configuration. The dashed lines show the ideal characteristics where the collector current is assumed to be constant in the active region.

The modulation in the neutral base width with V_{CB} was first investigated by James Early and is called **Early effect** after him. The dependence of I_C on V_{CB} can be modeled in the following way

$$\frac{\partial I_C}{\partial V_{CB}} = \frac{\partial I_C}{\partial W_B}\frac{\partial W_B}{\partial V_{CB}} \text{ …….(6.68)}$$

Since for a large reverse bias applied across C-B junction, I_C is approximately equal to $\alpha_F I_E$, the output conductance in common-base mode (g_{cb}) with constant emitter current can be expressed as

$$g_{cb} = \left.\frac{dI_C}{dV_{CB}}\right|_{I_E=\text{constant}} = I_E \frac{\partial \alpha_F}{\partial W_B}.\frac{\partial W_B}{\partial V_{CB}} \text{ …..(6.69)}$$

Assuming emitter efficiency $\gamma = 1$ and using eqn. (6.41) we can write $\alpha_F \approx \alpha_T \approx \left(1+\frac{W_B^2}{2L_n^2}\right)^{-1}$. For small values of W_B/L_n, g_{cb} can therefore be simplified as

$$\left|g_{cb}\right| \approx \frac{I_C W_B}{L_n^2}\frac{\partial W_B}{\partial V_{CB}} \text{ …..(6.70)}$$

For a large value of V_{CE}, $V_{CB} \approx V_{CE}$. Therefore, for the transistor in common-emitter mode, the output conductance (g_{ce}) with the (input) base current constant can be expressed as

$$g_{ce} = \left.\frac{dI_C}{dV_{CE}}\right|_{I_B=\text{constant}} \approx \left.\frac{dI_C}{dV_{CB}}\right|_{I_B=\text{constant}} \text{ ….(6.71)}$$

When $\gamma = 1$, it can be shown that $\beta \approx (2L_n^2/W_B^2)$ [Refer to eqn.(6.45)]. Using this relation and $I_C \approx \beta I_B$, from eqn.(6.71), we get

$$\left|g_{ce}\right| = I_B \frac{\partial \beta}{\partial W_B}\frac{\partial W_B}{\partial V_{CB}} = \frac{2I_C}{W_B}\frac{\partial W_B}{\partial V_{CB}} \text{ .....(6.72)}$$

Q. 6.8. Show that the output conductance in the common-base mode is smaller than that in the common-emitter mode.

From eqns. (6.70) and (6.72), it can be easily seen that the ratio of g_{cb} to g_{ce} is given by

$$\frac{g_{cb}}{g_{ce}} = \frac{W_B}{L_n^2}\frac{W_B}{2} = \frac{W_B^2}{2L_n^2} \approx \frac{1}{\beta}$$

As usually $\beta >> 1$, g_{cb} is $<< g_{ce}$.

From the above discussion, it can be seen that in common-emitter mode, the collector current increases appreciably with V_{CE}, instead of staying constant. The Early effect is modeled in circuit simulators with the help of a parameter called the **Early Voltage (V_A)**. Using eqn. (6.27), and assuming only a small change in base width, we may write

$$\frac{\partial I_C}{\partial V_{BC}} = \frac{\partial I_C}{\partial W_B}\frac{\partial W_B}{\partial V_{BC}} = -\frac{I_C}{W_B(0)}\frac{\partial W_B}{\partial V_{BC}} = -\frac{I_C}{V_A} \quad \text{.....(6.73)}$$

where $W_B(0)$ is the base width at $V_{BC} = 0V$ and

$$V_A = \left[\frac{1}{W_B(0)}\frac{\partial W_B}{\partial V_{BC}}\right]^{-1} \quad \text{.....(6.74)}$$

Now assuming that in the active region V_{BE} is fixed for a constant I_B, from eqn.(6.73) we can write

$$\frac{\partial I_C}{\partial V_{CE}} = -\frac{\partial I_C}{\partial V_{BC}} = \frac{I_C}{V_A} \quad \text{.....(6.75)}$$

From eqn. (6.75), we see that if the common-emitter characteristics in the active region were extrapolated, the intercept on the voltage axis would be at $-V_A$ (since V_A is usually much larger than the operating voltage V_{CE}) as shown in Fig.6.18.

Now, from eqn.(6.74), we have $\int_{W_B(0)}^{W_B} dW_B = \frac{W_B(0)}{V_A} \int_0^{V_{BC}} dV_{BC}$, or

$$W_B = W_B(0)\left[1 + \frac{V_{BC}}{V_A}\right] \text{.....(6.76)}$$

Now from eqn.(6.56), since I_S is inversely proportional to W_B, we have

$$I_S = \frac{I_S(0)}{1 + \frac{V_{BC}}{V_A}} \approx I_S(0)\left[1 - \frac{V_{BC}}{V_A}\right] \text{.....(6.77)}$$

If $\gamma \approx 1$ for the bipolar transistor, so that β_F is determined by α_T, then $\beta_F \propto (1/W_B)^2$. In that case

$$\beta_F = \frac{\beta_F(0)}{\left(1 + \frac{V_{BC}}{V_A}\right)^2} \approx \beta_F(0)\left(1 - \frac{2V_{BC}}{V_A}\right) \text{....(6.78)}$$

On the other hand, if $\alpha_T \approx 1$, so that β_F is determined by γ, then $\beta_F \propto (1/W_B)$. Therefore

$$\beta_F = \frac{\beta_F(0)}{\left(1 + \frac{V_{BC}}{V_A}\right)} \approx \beta_F(0)\left(1 - \frac{V_{BC}}{V_A}\right) \text{....(6.79)}$$

Eqns. (6.77) and (6.78) or (6.79) can now be used in Ebers-Moll model to incorporate the Early effect.

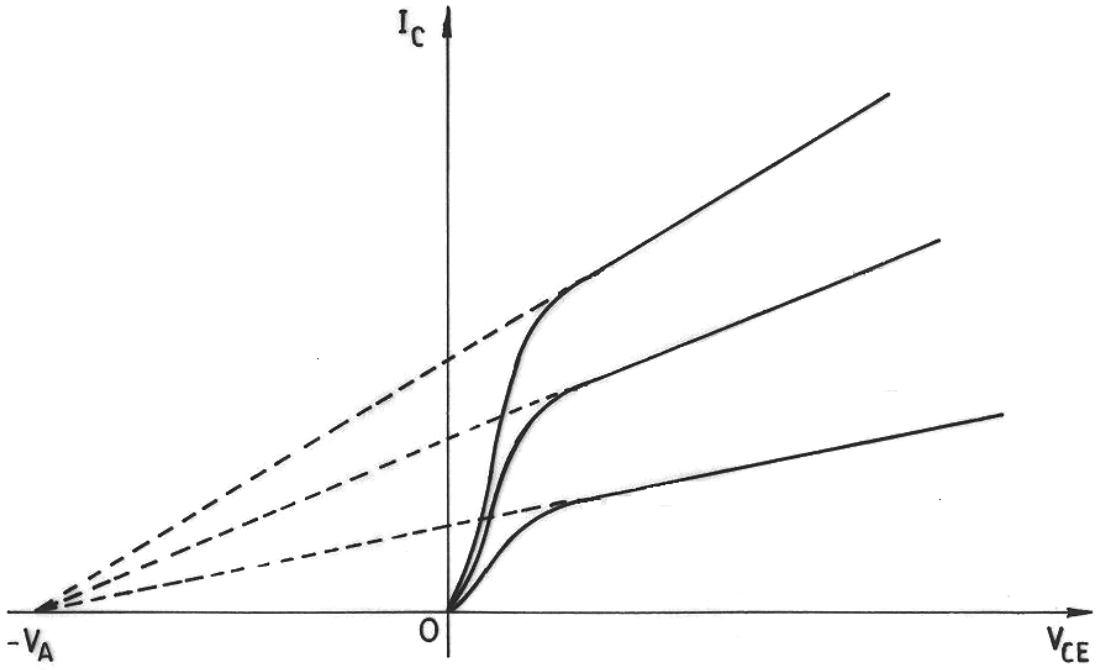


Fig.6.18: Graphical extrapolation of the I-V characteristics to extract the value of Early voltage (V_A)

6.9 Limitation on the Junction Voltage

If the reverse bias applied to the C-B junction is continuously increased, the BJT ultimately 'breaks down' and the collector current increases sharply as shown in Fig. 6.19. This breakdown may be either due to base punch-through or avalanche multiplication at the C-B junction.

From the discussion in the previous section, it is evident that as V_{CB} is increased, the neutral base width decreases. Finally, for a particular value of V_{CB}, the depletion region at C-B junction extends all the way up to E-B junction and the neutral base width becomes zero. This is called **base punch-through**. At punch-through, the collector current increases sharply as the electrons are swept directly from the emitter into the collector and the transistor action is lost. This is a breakdown condition and is to be avoided. For an n^+pn^+ transistor, assuming that the C-B depletion region exists mostly in

the base, the voltage applied across C-B junction at punch-through can be approximated as

$$V_{PT} = \frac{qN_A W_B^2}{2\varepsilon_s} \text{.....(6.80)}$$

In most cases, however, avalanche breakdown of the collector-base junction occurs before the punch-through condition is reached. Interestingly, it is found that the **breakdown voltage in common-emitter mode (V_{CEO})** is considerably smaller than that in the **common-base mode (V_{CBO})**. Let us now analyze these two situations.

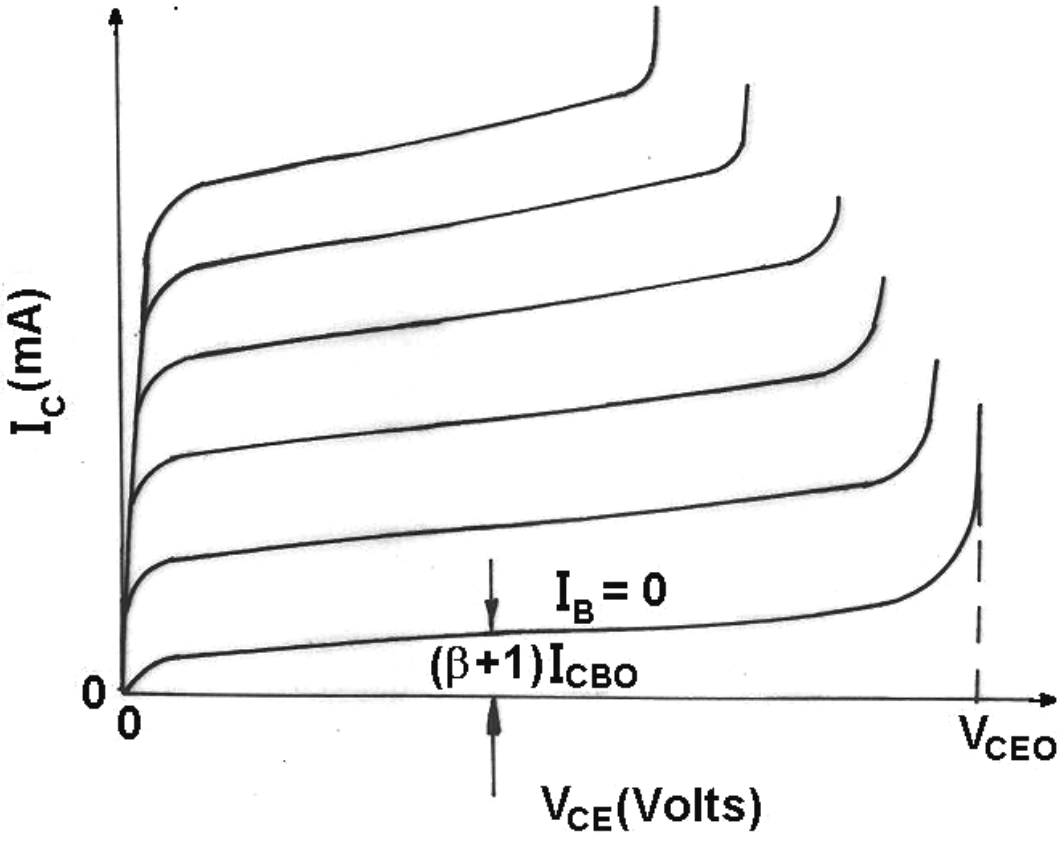


Fig. 6.19: Common emitter output characteristics of a BJT showing breakdown

In the presence of a large reverse voltage applied across C-B junction, the current flowing out of the collector-base depletion region is multiplied by the collector multiplication factor M. Thus, in such a situation the collector current given by eqn. (6.31) is modified as

$$I_C = (\alpha I_E + I_{CBO})M \quad \text{.....(6.81)}$$

As in eqn.(4.87), M can be expressed as

$$M = \frac{1}{1 - \left(\frac{V_{CB}}{V_{BR}}\right)^m} \quad \text{……(6.82)}$$

where m is a constant. Eliminating M from eqns. (6.81) and (6.82), we get

$$V_{CB} = \left[1 - \frac{I_{CBO} + \alpha I_E}{I_C}\right]^{1/m} V_{BR} \quad \text{……..(6.83)}$$

From eqn.(6.83), we see that for the common-base configuration, the maximum value of V_{CB} is obtained in the limiting case of $I_E = 0$. This value of voltage across the C-B junction (V_{CBO}) with emitter open (i.e. $I_E = 0$) is given as

$$V_{CBO} = \left[1 - \frac{I_{CBO}}{I_C}\right]^{1/m} V_{BR} \approx V_{BR} \quad \text{……(6.84)}$$

However, for a nonzero I_E, the maximum voltage that can be applied across the C-B junction without causing avalanche multiplication of carriers will be less than V_{CBO}.

On the other hand, in the common-emitter configuration, the current amplification in addition to the avalanche multiplication imposes additional restriction on the maximum voltage that can be applied across C-B junction. It can be readily seen from eqn. (6.81) that if the input current (I_B) in common-emitter configuration is made zero, so that $I_E = I_C$, we get

$$I_C = \frac{MI_{CBO}}{1-\alpha M} \text{ ……..(6.85)}$$

This means that I_C will increase indefinitely if $\alpha M = 1$. Since α is only slightly less than 1, for this condition it only needs M to be slightly greater than 1. By contrast, for the common-base configuration, if the input current I_E is zero, M needs to be infinitely high for the collector current to increase in a similar manner. Thus the maximum collector-emitter voltage that can be applied with base open (V_{CEO}) is much less than V_{CBO}.

Let us now obtain an expression for V_{CEO}. From eqn. (6.83), noting that $\alpha = \beta/(1+\beta)$, we can write

$$V_{CB} = \left[1 - \frac{\beta}{1+\beta}\left(1 + \frac{I_B}{I_C}\right) - \frac{I_{CBO}}{I_C}\right]^{1/m} V_{BR} \text{ …..(6.86)}$$

For a large value of V_{CE}, V_{CB} is approximately equal to V_{CE}. Thus in the limiting case, for $I_B = 0$, we get

$$V_{CEO} \approx \frac{V_{BR}}{(1+\beta)^{1/m}} \text{ …..(6.87)}$$

Comparing eqns. (6.84) and (6.87), we can easily verify that $V_{CEO} < V_{CBO}$.

Example 6.3: For a bipolar transistor $\beta = 80$. Assuming $m = 4$, find out the ratio of V_{CBO} to V_{CEO}.

Solution: From eqn. (6.87), we get $V_{CEO} \approx \frac{V_{BR}}{\sqrt[4]{81}} = \frac{V_{BR}}{3}$, while $V_{CBO} \approx V_{BR}$.

Therefore, $V_{CBO}/V_{CEO} = 3$

In most bipolar transistors, the doping concentration in the collector is less than in the base. This helps to prevent base punch-through and reduce Early Effect, as the depletion width now extends more into the collector rather than into the base. Also, with a lower collector doping concentration, V_{BR} increases (refer to eqn.(4.88)), and therefore the device can withstand a larger collector bias voltage.

6.10 Capacitances in a BJT

Fig. 6.9 shows the presence of stored charges in the various regions of the BJT. As in the case of diodes discussed in Section 4.43, these charges give rise to a capacitive effect. We shall now develop a model for these capacitances.

Let us first consider $V_{BE} \neq 0$, $V_{BC} = 0$. For this condition, in Fig. 6.9, $Q_{BR} = Q_C = 0$, and the stored charges are given by Q_E and Q_{BF}. Now, from the areas of the shaded regions representing the charges, we can write

$$Q_E = \frac{Aqp'_{ne}W_E}{2} = \frac{Aqn_i^2 W_E}{2N_{DE}}\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right] \ldots.(6.88)$$

and

$$Q_{BF} = \frac{Aqn'_{pe}W_B}{2} = \frac{Aqn_i^2 W_B}{2N_A}\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right] \ldots..(6.89)$$

Now, considering the expression for I_C given by eqn.(6.55), we can write

$$Q_F = Q_E + Q_{BF} = \tau_F I_C \ldots.(6.90)$$

where τ_F is the **forward transit time** given by

$$\tau_F = \frac{\frac{W_E}{2N_{DE}} + \frac{W_B}{2N_A}}{\frac{D_n}{N_A W_B}} = \frac{N_A W_B W_E}{2N_{DE}D_n} + \frac{W_B^2}{2D_n} \ldots\ldots(6.91)$$

We can see that τ_F is a constant for a particular device structure and is independent of bias. We can now define the **diffusion capacitance for the E-B junction (C_{DE})** from eqn.(6.90) and (6.55) as

$$C_{DE} = \frac{dQ_F}{dV_{BE}} = \tau_F \frac{dI_C}{dV_{BE}} \approx \tau_F \frac{d\left[I_S \exp\left(\frac{V_{BE}}{V_T} \right) \right]}{dV_{BE}} = \frac{\tau_F I_{CC}}{V_T} \quad \text{.....(6.92)}$$

Similarly, considering the reverse operation, i.e. $V_{BE} = 0$, $V_{BC} \neq 0$, and following the same procedure as above, we can obtain an expression for the **diffusion capacitance of the C-B junction (C_{DC})** as

$$C_{DC} = \frac{\tau_R I_{EC}}{V_T} \quad \text{.....(6.93)}$$

where τ_R is the **reverse transit time** of the transistor.

In addition to the to the stored mobile charges in the neutral regions, which gives rise to the diffusion capacitances, there are the fixed charges in the E-B and C-B depletion regions, which give rise to the junction capacitances. The expressions for the junction capacitances are similar to that for a diode given by eqn. (4.31). Therefore the expressions for the **E-B and C-B junction capacitances denoted by C_{JE} and C_{JC}** respectively are given by

$$C_{JE} = C_{JEO} \left(1 - \frac{V_{BE}}{V_{biE}} \right)^{-m_E} \quad \text{.....(6.94)}$$

and

$$C_{JC} = C_{JCO} \left(1 - \frac{V_{BC}}{V_{biC}} \right)^{-m_C} \quad \text{.....(6.95)}$$

where C_{JE0} and C_{JC0} are the values of the E-B and C-B junction capacitances at $V_{BE} = 0$ and $V_{BC} = 0$, V_{biE} and V_{biC} are the built-in potentials of the E-B and C-B junctions and m_E and m_C are the factors dependent on the nature of grading of E-B and C-B junctions.

The complete Ebers-Moll equivalent circuit, including capacitances, is shown in Fig.6.20.

6.11 Switching of Bipolar transistors

In digital circuits, the bipolar transistor is often used as an electrically controllable switch, which can respond to input changes much more rapidly than a mechanical switch. This will be clear from the circuit shown in Fig. 6.21(a). In Fig.6.21(b), the output characteristics of the transistor connected in the common-emitter configuration, together with the load line is shown. Fig.6.21(c) shows a circuit where the bipolar transistor has been replaced by a switch. When the input voltage (v_{in}) is low (almost 0V or less) in Fig. 6.21(a), the transistor is cut-off and the operating point is shown by point A in Fig. 6.21(b). At this point, $I_C \approx 0$ and $v_{out} = V_{CE} \approx V_{CC}$. The situation corresponds to the switch in Fig. 6.21(c) being in the open condition. On the other hand, when v_{in} is increased so that I_B is sufficiently high, the operating point moves to point B in the saturation region of the characteristics in Fig. 6.21(b). Since $I_C < \beta I_B$ in the saturation region, the minimum base current required for the transistor to go to saturation is $\left(I_{Bsat}\right)_{min} = \frac{I_{Csat}}{\beta} = \frac{V_{CC} - V_{CEsat}}{\beta R_L}$. At this point $v_{out} = V_{CE} = V_{CEsat}$ and $I_C = I_{Csat} = (V_{CC} - V_{CEsat})/R_L$. Since V_{CEsat} is small (≈ 0.2V), we can write $I_C \approx V_{CC}/R_L$ and $v_{out} \approx 0$. This is similar to the situation when the switch is closed in Fig. 6.21(c). Thus we see that in Fig. 6.21(a), the transistor behaves as a switch and the flow of current through a load resistance can be controlled by a much smaller input current (since $I_B << I_C$).

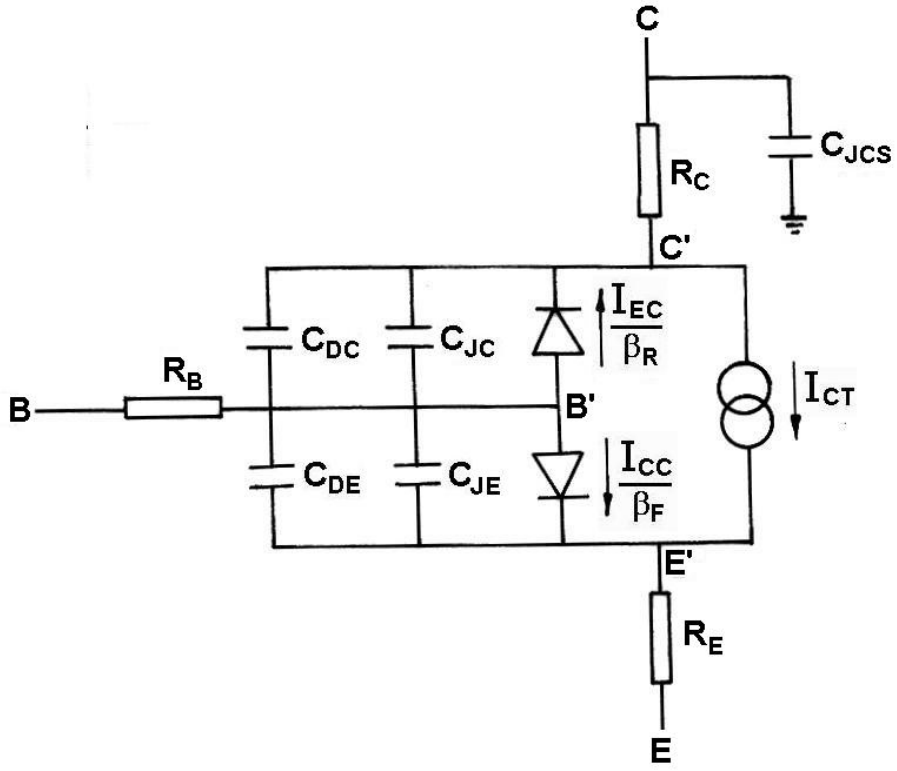


Fig.6.20: Ebers-Moll equivalent circuit of a BJT including capacitances

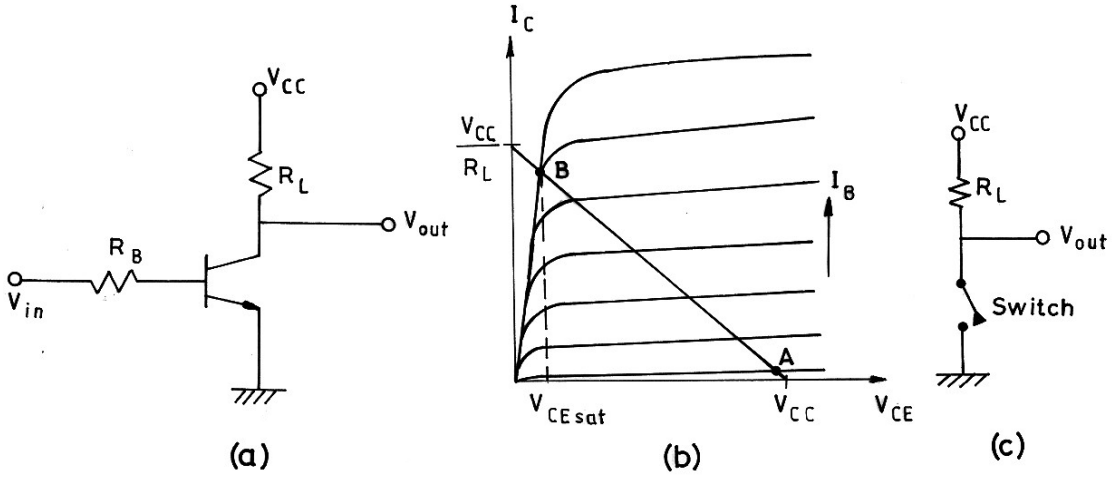


Fig. 6.21 (a) A circuit using BJT as a switch (b) common emitter output characteristics of BJT with load line and (c) an equivalent circuit where the BJT is replaced by a switch

Note that the given circuit also performs the operation of an inverter, which is a basic component in a digital circuit. We have already seen that when v_{in} is LOW, $v_{out} \approx V_{CC}$ (HIGH). On the other hand, when v_{in} is sufficiently HIGH, $v_{out} = V_{CEsat}$ (LOW). Although the circuit is useful as an inverter, an important aspect of its operation is the delays involved in switching from one state to the other. We shall now discuss this in detail.

Let us consider the circuit shown in Fig. 6.22(a). The input waveform, as shown in Fig. 6.22(b), changes from $v_{in} = -V_R$ to $v_{in} = V_F$ at $t = 0$ and remains at that value till $t = t_1$, when v_{in} again changes to $-V_R$. At $t = 0^-$, the transistor was in cut-off condition with $V_{BE} = -V_R$, $I_B = 0$ and the excess minority carrier concentration in the base $Q_{nB} \approx 0$ (the small negative value is neglected) as shown in Figs. 6.22(c), (d) and (e) respectively. At $t = 0^+$, the change in the input voltage to V_F results in a forward base current. Consequently, V_{BE} changes quickly from $-V_R$ to 0, but then increases to a small positive value, as the stored charge builds up in the base. The situation is similar to the switching of a diode as discussed in Section 4.4. The base current in the entire interval $0 < t < t_1$ is almost a constant and is given by $I_{FB} = \frac{V_F - V_{BE}}{R_B} \approx \frac{V_F}{R_B}$. For this period, considering only the charges stored only in the base (i.e. for $\gamma = 1$), we can write

$$\frac{dQ_{nB}}{dt} = I_{FB} - \frac{Q_{nB}}{\tau_n} \quad \ldots\ldots(6.96)$$

Eqn.(6.96) is another form of eqn.(4.62) and essentially states that the rate of increase of base charge is equal to the difference between the rate at which charges flow in (I_{FB}) and the rate at which they recombine $\left(\frac{Q_{nB}}{\tau_n}\right)$. The solution of eqn. (6.96) is given by

$$Q_{nB}(t) = I_{FB}\tau_n\left[1-\exp\left(-\frac{t}{\tau_n}\right)\right] \text{ …….(6.97)}$$

Thus the base charge increases with time, and the collector current also increases according to the relation $I_C = \frac{Q_{nB}}{\tau_{tB}}$ (from eqn. (6.29)) as shown in Fig. 6.22(f), where τ_{tB} is the base transit time. However, at $t = t_{on}$, I_C reaches its saturation value I_{Csat}. The corresponding charge is given by $Q_{nB} = I_{Csat}\tau_{tB}$. Therefore, substituting $t = t_{on}$ and $Q_{nB}(t) = I_{Csat}\tau_{tB}$ in eqn. (6.97), and using eqn. (6.45), we have

$$t_{on} = \tau_n \ln\frac{1}{1-\frac{I_{Csat}}{\beta I_{FB}}} \text{ …..(6.98)}$$

We can see from eqn.(6.98) that a higher value of I_{FB} reduces the delay time. For example, if $I_{FB} = 2I_{Csat}/\beta$, $t_{on} = \tau_n\ln(2)$, while if $I_{FB} = 5I_{Csat}/\beta$, $t_{on} = \tau_n\ln(1.25)$.

Although the collector current is limited by the circuit, the base charge continues to increase following eqn.(6.97), and if t_1 is sufficiently long, will reach its ultimate value of $Q_{nB} = I_{FB}\tau_n$. This is shown in Fig.6.22(e). At $t = t_1$, when v_{in} changes to $-V_R$, the direction of the base current changes. As carriers flow out of the base, the stored charge in the base reduces. However, V_{BE}, which is related to the stored charge in the base, reduces slowly and reaches 0V at $t = t_1 + t_{off}$. In the period $t_1 < t < t_1 + t_{off}$, the base current is almost constant at $-I_{RB} = \frac{-V_R - V_{BE}}{R_B} \approx -\frac{V_R}{R_B}$. Beyond $t = t_1 + t_{off}$, V_{BE} quickly falls to $-V_R$ and then $I_B = 0$. Now, in the period $t_1 < t < t_1 + t_{off}$, the decay of the stored charge in the base is governed by the relation

$$\frac{dQ_{nB}}{dt} = -I_{RB} - \frac{Q_{nB}}{\tau_n} \text{ ……(6.99)}$$

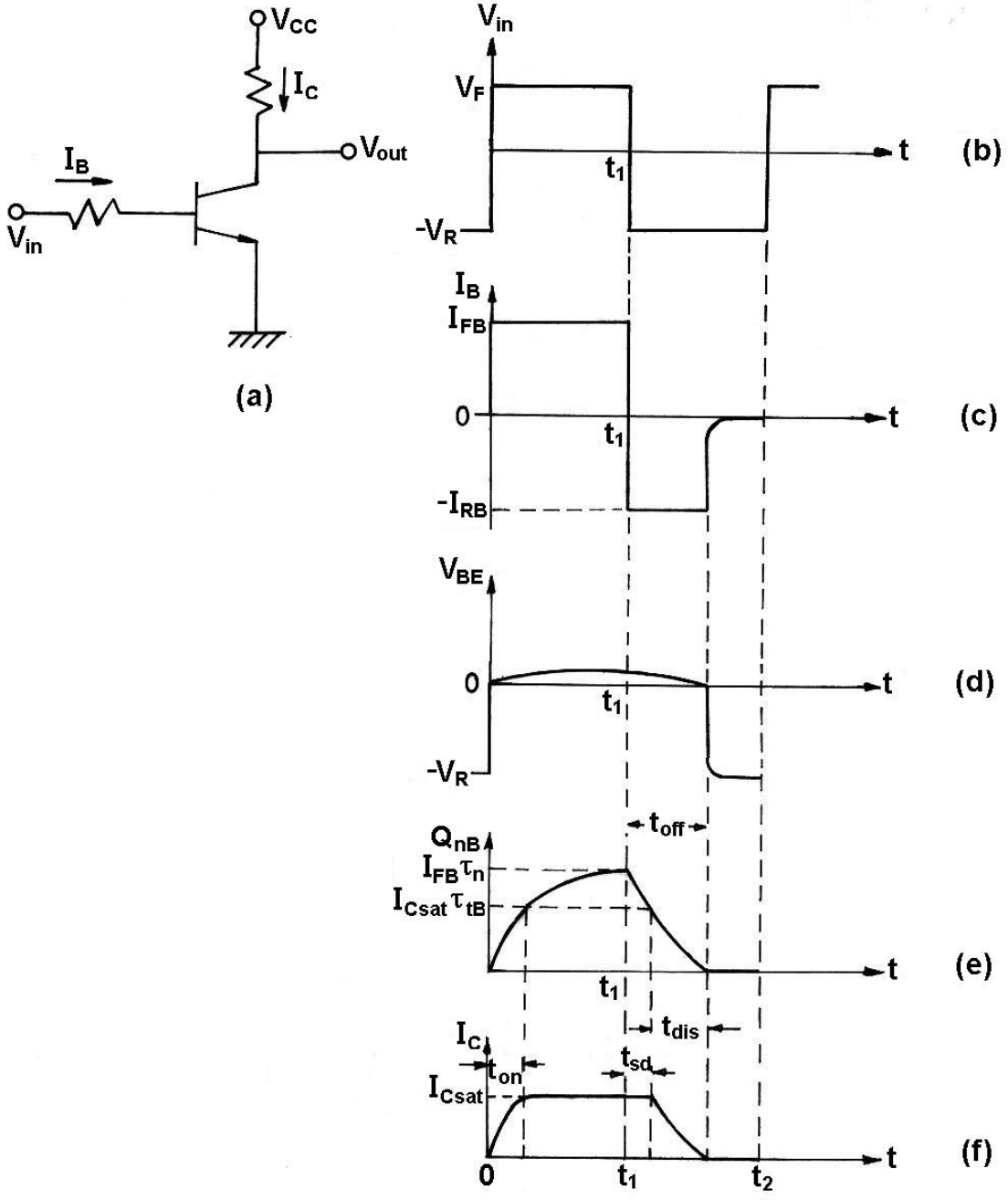


Fig.6.22: (a) An inverter circuit using a BJT; (b) The input voltage waveform; (c) Variation in V_{BE}; (d) Variation in I_B; (e) Excess minority charge variation in base ; (f) Variation in I_C , showing the different time delays associated with switching of the BJT.

From eqn.(6.99), we see that $\frac{dQ_{nB}}{dt}$ is negative, and the rate of decay of the charges is equal to the sum of the rate at which carriers are flowing out (I_{RB}) and the rate at which they are recombining $\left(\frac{Q_{nB}}{\tau_n}\right)$. Considering $Q_{nB} = I_{FB}\tau_n$ at $t = t_1$, the solution of eqn.(6.99) is given by

$$Q_{nB}(t) = I_{FB}\tau_n \exp\left(-\frac{t-t_1}{\tau_n}\right) - I_{RB}\tau_n\left[1-\exp\left(-\frac{t-t_1}{\tau_n}\right)\right] \ldots\ldots\ldots(6.100)$$

As the stored charge decays, the transistor comes out of saturation and I_C starts to reduce only when $Q_{nB} = I_{Csat}\tau_{tB}$. The duration for which the transistor remains in saturation even after the input voltage has been reduced to a negative value is called the **storage delay time (t_{sd})**. Therefore substituting $Q_{nB} = I_{Csat}\tau_{tB}$ and $t = t_1 + t_{sd}$ in eqn.(6.100), we get

$$t_{sd} = \tau_n \ln \frac{I_{RB} + I_{FB}}{I_{RB} + \frac{I_{Csat}}{\beta}} \ldots\ldots\ldots(6.101)$$

From eqn. (6.101), we see that t_{sd} reduces with increases with increase in I_{RB}. This is to be expected, as a larger reverse base current will result in a faster reduction in base charge. Now, at $t = t_1 + t_{off}$, $Q_{nB} = 0$, and therefore an expression for t_{off} can be obtained by substituting these values in eqn.(6.100). The discharge time (t_{dis}) is defined as the time required for the transistor to go from saturation to cut-off and is given by $t_{dis} = t_{off} - t_{sd}$. An expression for t_{dis} can now be obtained as

$$t_{dis} = \tau_n \ln\left(1 + \frac{I_{Csat}}{\beta I_{RB}}\right) \ldots\ldots(6.102)$$

In all the expressions given by eqns.(6.98), (6.101) and (102), we find that the delays are proportional to τ_n. Therefore, in switching transistors, the minority carrier lifetime in the

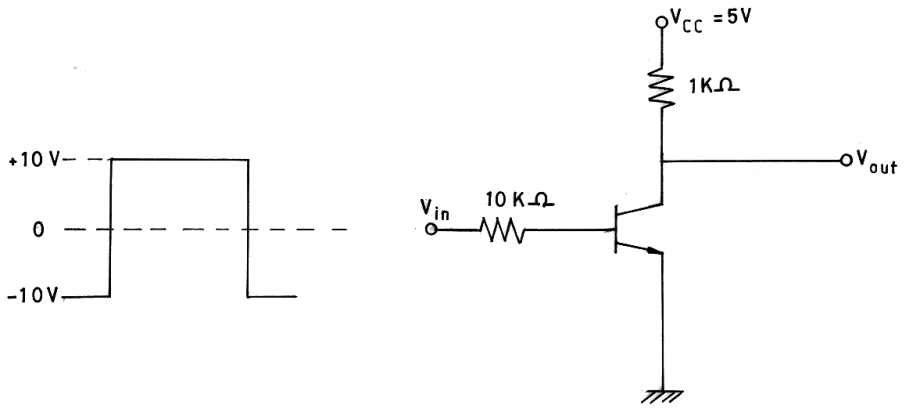
base must be made as low as possible. One way of reducing the lifetime is by doping with impurities, such as gold, which give rise to energy level near the middle of the bandgap. However, with reduction in τ_n, the β of the transistor reduces (refer to eqn.(6.45). To overcome this, the base transit time should also be reduced as far as possible, by reducing the base width.

It may be mentioned here that another component of delay, which is due to the charging of the junction capacitances, has been neglected in the above analysis.

Q 6.9 Why does it take a long time for V_{BE} to change by a small voltage when the E-B junction is forward biased, while V_{BE} changes almost instantaneously by a large voltage when the E-B junction is reverse biased?

Let the doping concentration in the base of an n-p-n silicon bipolar transistor be $10^{17}/cm^3$. Therefore the minority carrier concentration at thermal equilibrium at room temperature in the base is given by $n_{po} = 2.25x10^3/cm^3$. Now, when $V_{BE} = 0.6V$, $n_{pE} = n_{po}\exp\left(\frac{V_{BE}}{V_T}\right) = 2.37x10^{13}/cm^3$, while when $V_{BE} = -5V$, $n_{pE} \approx 0$. Thus, to change V_{BE} from 0V to 0.6V, n_{pE}, and therefore Q_{nB}, has to change by many orders of magnitude more than when V_{BE} changes from 0 to –5V. This explains why the change in V_{BE} when the E-B junction is forward biased is a much slower process.

Example 6.4 In the figure shown below, a square wave is applied as input to switch the BJT between ON and OFF states. Neglecting the effect of the junction capacitance, the total delay in one cycle may be given by $t_{delay} = t_{on} + t_{sd} + t_{dis}$. Calculate (a) t_{on}/t_{delay}, (b) t_{sd}/t_{delay} and (c) t_{dis}/t_{delay}. Assume $V_{CEsat} = 0.2V$ and $\beta = 50$.



From the above figure, $I_{Csat} = \dfrac{V_{CC} - V_{CEsat}}{R_C} = \dfrac{5-0.2}{1}\text{mA} = 4.8\text{mA}$

$I_{FB} \approx \dfrac{V_F}{R_B} = \dfrac{10}{10}\text{mA} = 1\text{mA}$ and $-I_{RB} \approx \dfrac{-V_R}{R_B} = \dfrac{-10}{10}\text{mA} = -1\text{mA}$

Now, from eqn.(6.98), $t_{on} = \tau_n \ln \dfrac{1}{1 - \dfrac{I_{Csat}}{\beta I_{FB}}} = \tau_n \ln \dfrac{1}{1 - \dfrac{4.8}{50x1}} = 0.1\tau_n$

From eqn.(6.101), $t_{sd} = \tau_n \ln \dfrac{I_{RB} + I_{FB}}{I_{RB} + \dfrac{I_{Csat}}{\beta}} = \tau_n \ln \dfrac{1+1}{1 + \dfrac{4.8}{50}} = 0.60\tau_n$

From eqn.(6.102), $t_{dis} = \tau_n \ln\left(1 + \dfrac{I_{Csat}}{\beta I_{RB}}\right) = \tau_n \ln\left(1 + \dfrac{4.8}{50x1}\right) = 0.09\tau_n$

Now, $t_{delay} = (0.1+0.6+0.09)\tau_n = 0.79\tau_n$

Therefore (a) t_{on}/t_{delay} = (0.1/0.79) = 0.126, (b) t_{sd}/t_{delay} = (0.6/0.79) = 0.759 and (c) t_{dis}/t_{delay} = (0.09/0.79) = 0.114

We can see from the results that the storage delay time is by far the largest component of delay

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6.12 Process flow for an n-p-n bipolar junction transistor in Integrated Circuit

We shall now discuss the process flow to realize a bipolar junction transistor in an integrated circuit. This is schematically presented in Fig. 6.23. The starting material is p-type silicon having a resistivity of approximately 5 Ohm.cm.

Step1: Thermal oxide of thickness 0.5 μm is grown on the substrate.

Step2: Photolithography is carried out to open a window in the oxide.

Step3: Arsenic or antimony is ion-implanted (or diffused) through this window to form an n^+ region (Fig.6.23a)

Step4: SiO_2 is removed and an n-epitaxial layer is grown on the entire surface (Fig. 6.23b)

Step5: Unlike in the case of discrete devices, a transistor in an integrated circuit has to be properly isolated from adjacent transistors for the circuit to function. In this particular example, a reverse-biased p-n junction is used to isolate the transistors. The isolation regions completely surround the n^+ buried layer to form islands of n-type silicon as shown in Fig. 6.23(c). This is achieved by ion-implantation followed by a high-temperature drive-in to push the boron all the way through the epitaxial layer.

Step6: The base region is now defined by photolithography. A boron implant followed by drive-in is used to produce a base (Fig. 6.23d).

Step7: Emitter and collector contact regions are next defined by photolithography and phosphorus implantation is carried out. The n^+ diffusion in the collector is needed to produce a low-resistance ohmic contact (Fig. 6.23e).

Step8: The contact windows are opened over emitter, base and collector as shown in Fig.6.23(f).

Step9: Aluminium is evaporated over the entire surface. Afterwards, aluminium interconnections between the components of the circuit are defined. The final structure is shown in Fig.6.23(g).

Along with the npn transistor, a p-n junction diode is also realized in Fig. 6.23. The base doping for the transistor is used to realize the p-n junction. The diode is isolated from the transistor by the p-tubs realized in step 5. The impurity concentration profile in the emitter, base and collector of the transistor is shown in Fig. 6.24.

However, for modern transistors in high-speed digital circuits, the process flow is considerably modified. The main features include oxide isolation instead of junction isolation and use of doped polysilicon to realize emitter and base regions. These process modifications result in high packing density as well as reduction of parasitic sidewall capacitance and base resistance. The importance of these parameters will be discussed in the next chapter. The cross-sectional diagram of such a transistor with oxide isolation, polysilicon emitter and base is shown in Fig.6.25.

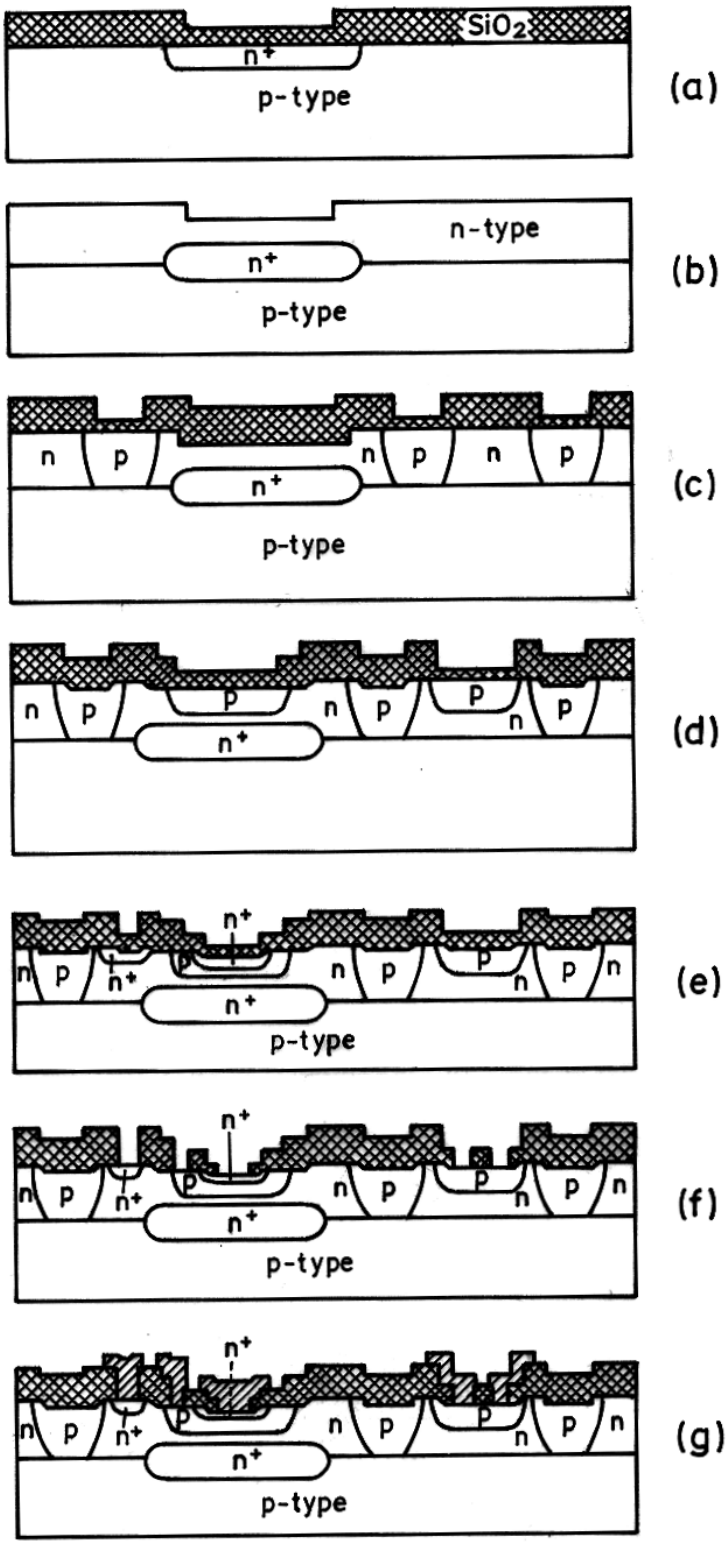


Fig. 6.23 : Process flow for an npn BJT

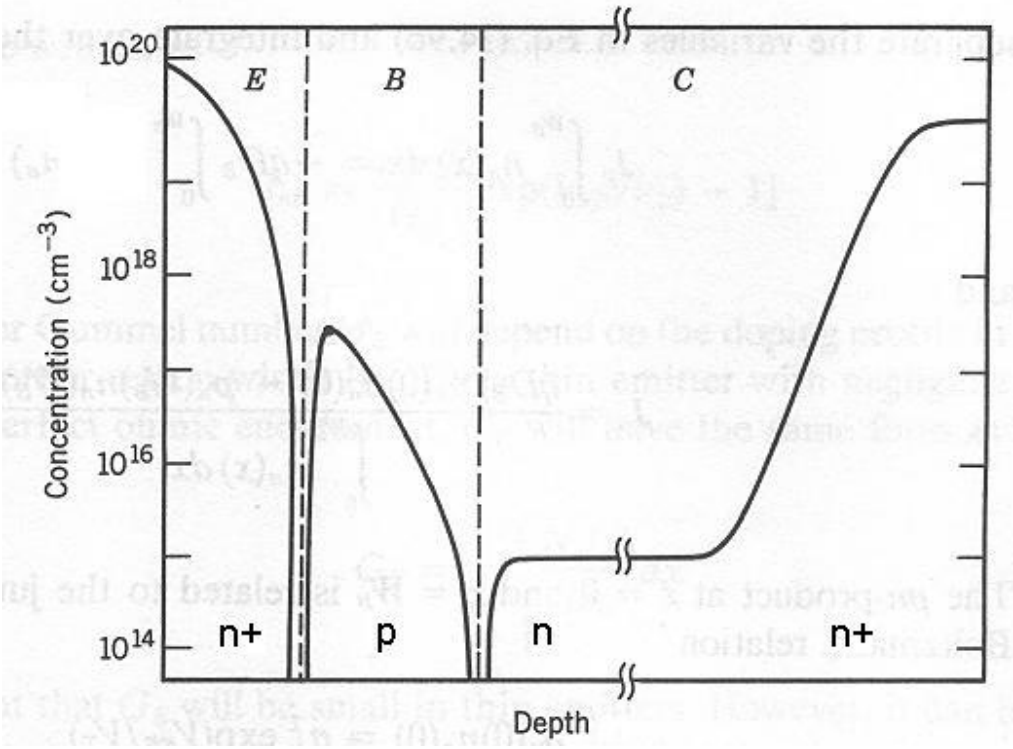


Fig. 6.24 The impurity concentration profile in the emitter, base and collector of the transistor

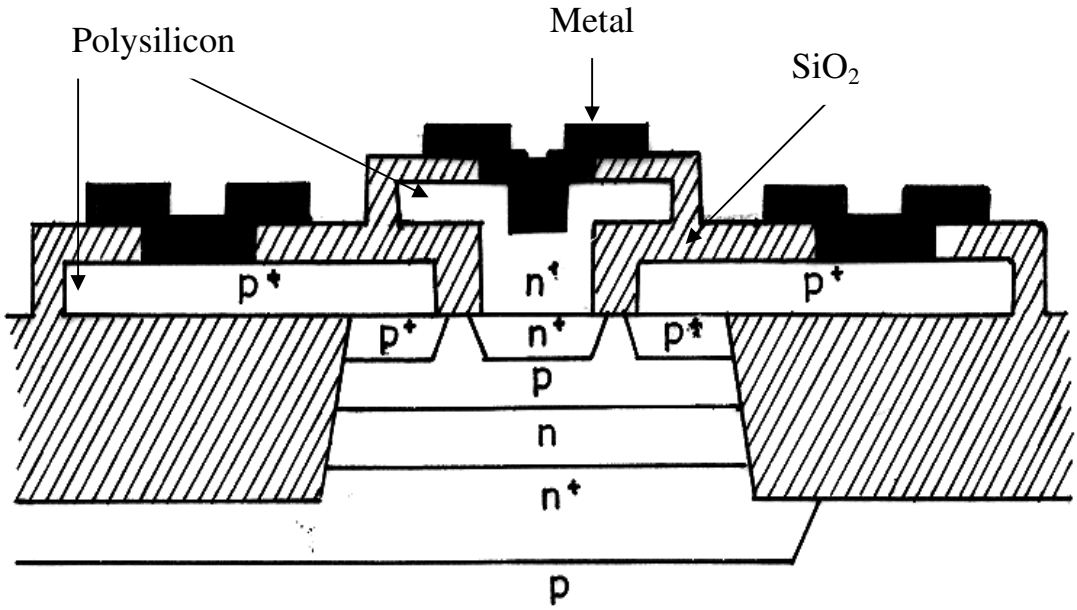


Fig.6.25. The cross-sectional diagram of an improved bipolar transistor with oxide isolation, polysilicon emitter and base

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PROBLEMS AND QUESTIONS

1. The β of an npn transistor is estimated to be 200 considering only the effect of γ and assuming $\alpha_T = 1$. On the other hand, considering only the effect of α_T and assuming $\gamma = 1$, the β of the same transistor is found to be 100. What is the actual β of the transistor if both α_T and γ are taken into account?

2. Two transistors T_1 and T_2 are identical in all respects except in their base widths. The emitter efficiencies are unity in both cases. The base width of T_1 is 1μm and that of T_2 is 2μm. The transistors are biased in the active region of operation. The base current of T_1 is 10μA when the collector current is 1mA. Determine the base current of T_2 when the collector current is 1mA.

3. The β of an n^+pn^+ transistor is 100 when the collector-base junction is reverse biased with $V_{CB} + V_{bi} = 1V$. The neutral base width, i.e. the undepleted portion of the base for this case is 1.2μm and the depletion layer width at the C-B junction is 0.3 μm. Determine the β of the transistor when $V_{CB} + V_{bi} = 9V$. Assume $\alpha_T = 1$ and note that V_{CB} is the reverse voltage applied at the C-B junction while V_{bi} is the built-in potential of this junction.

4. The BJT T1, whose parameters are given in Example 6.1, has an area $A=10^{-3}cm^2$. The emitter-base junction of T1 is forward-biased and the collector base junction is left open. If the hole current injected into the emitter is $I_{pE} = 10\ \mu A$,

   (i) Determine the voltage at the collector-base junction.

   (ii)Sketch the minority carrier distribution in the base

   (iii) Calculate the value of the emitter current.

(iv) If now the collector-base junction is shorted, without changing the voltage at the E-B junction, what will be the value of the collector current assuming W_B does not change?

(v) Sketch the minority carrier distribution in the base for case (iv)

(vi) Calculate the base current for case (iv)

5. In an n^+pn^+ transistor with uniform base doping, the neutral base width (i.e. the undepleted portion of the base) is 1.2 μm when V_{CB} = 5V. The depletion layer width at the collector-base junction for this case is 0.3 μm. Find the value of the collector-base voltage at which the entire base gets depleted. Neglect the built-in potential as well as the depletion layer width at the emitter-base junction.

6. In an n^+pnn^+ transisistor, the n^+pn regions are identical to the BJT T1, whose parameters are given in Example 6.1. Assuming that avalanche breakdown at the C-B junction does not occur before base punchthrough, (a) calculate the base punchthrough voltage (V_{PT}). (b) What is the maximum electric field at the C-B junction when $V_{CB} = V_{PT}$? (c) If the critical electric field for avalanche breakdown is 20V/μm, what is the maximum possible value of V_{CB} for this transistor?